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(54) **MULTIPLIER SUPPORTING VARIOUS PRECISIONS AND DATATYPES AND OPERATING METHOD THEREOF**

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G06F 7/523 (2006.01)

(52) **U.S. Cl.**
CPC ***G06F 7/523*** (2013.01)

(57) **ABSTRACT**

An electronic device includes a multiplier including a multiplication logic, a memory including at least one instruction, and at least one processor configured to execute the at least one instruction and to obtain a first input value and a second input value, identify datatypes and precisions of the first input value and the second input value, based on the identified datatypes and precisions, distribute bits of the first input value and bits of the second input value to sub-multiplication logics of the multiplication logic, and obtain at least one output of the multiplication logic based on outputs of the sub-multiplication logics.

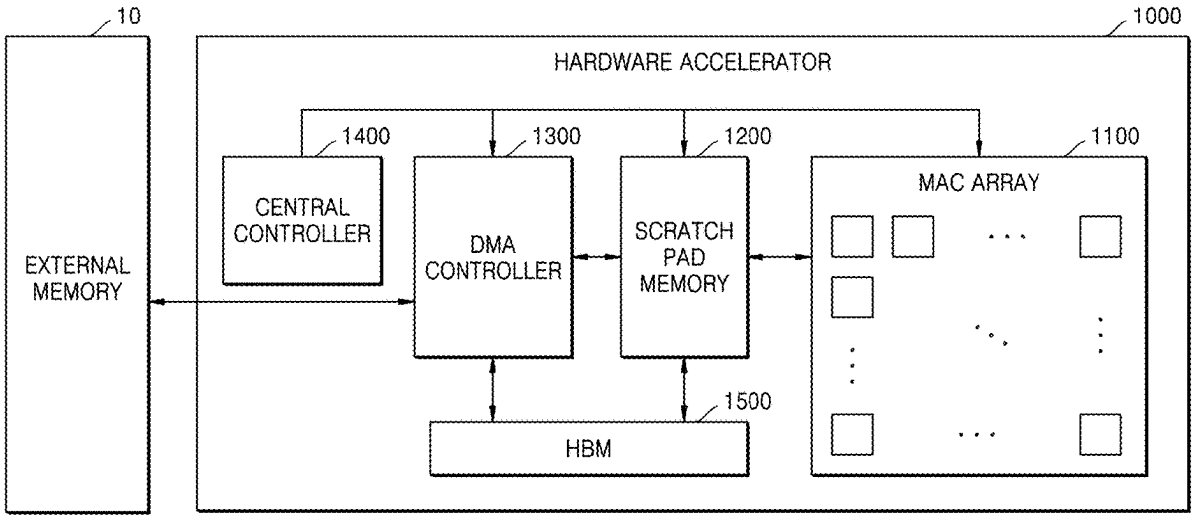


FIG. 1

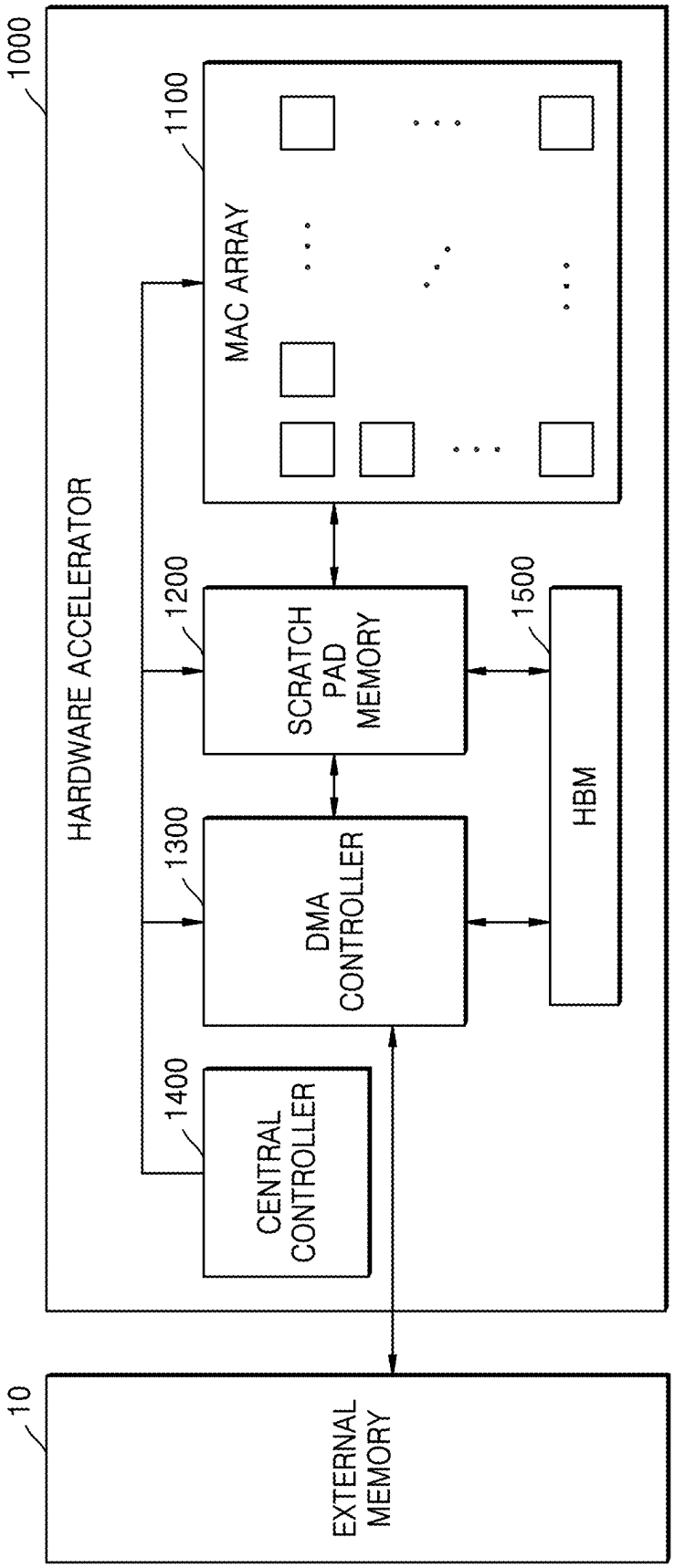

1000
HARDWARE ACCELERATOR
1100
MAC ARRAY
1200
SCRATCH PAD MEMORY
1500
HBM
1300
DMA CONTROLLER
1400
CENTRAL CONTROLLER
10
EXTERNAL MEMORY

FIG. 2

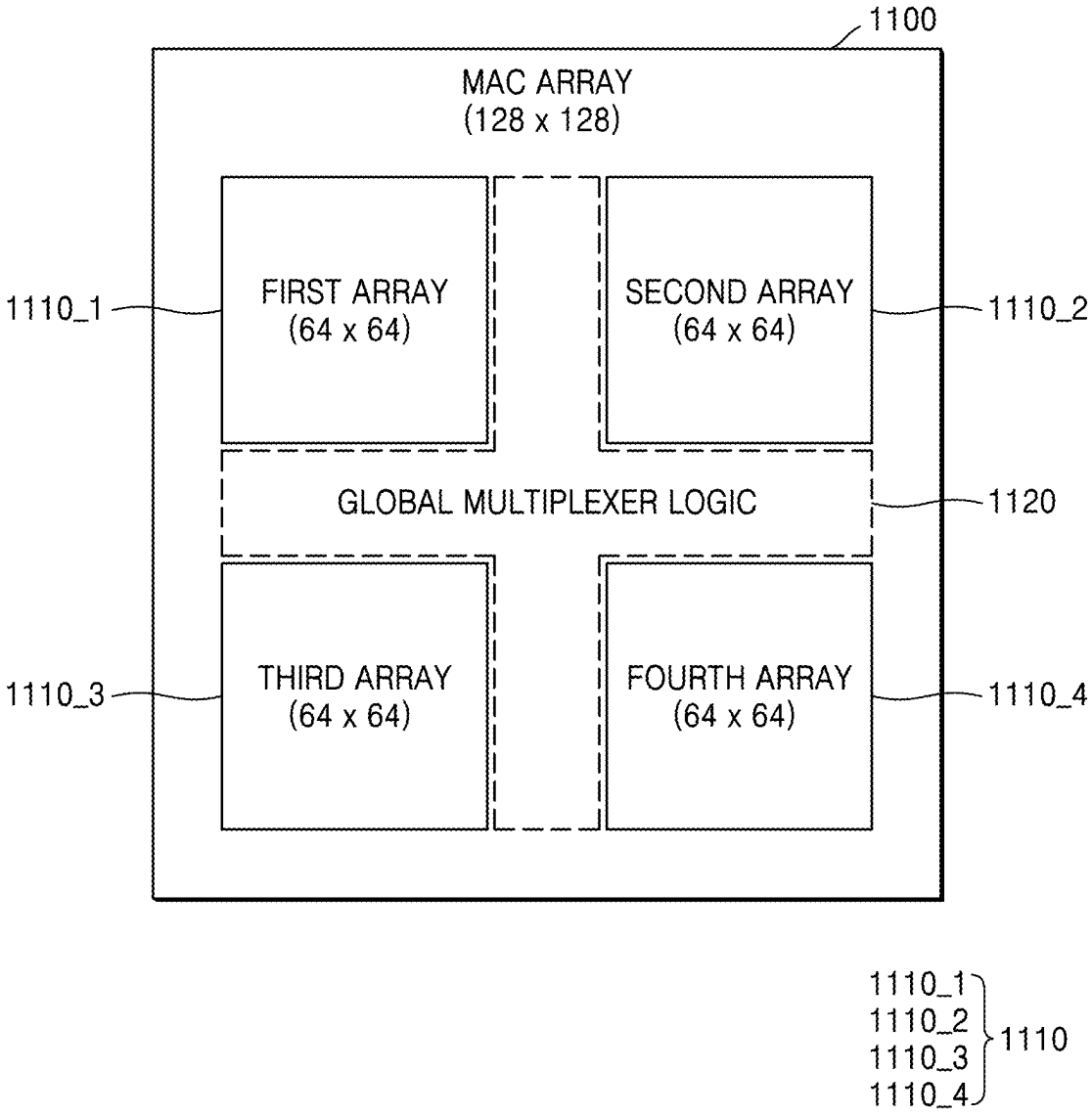

1100
MAC ARRAY
(128 x 128)
FIRST ARRAY
(64 x 64)
SECOND ARRAY
(64 x 64)
GLOBAL MULTIPLEXER LOGIC
THIRD ARRAY
(64 x 64)
FOURTH ARRAY
(64 x 64)
1110_1
1110_2
1110_3
1110_4
1120
1110

FIG. 3

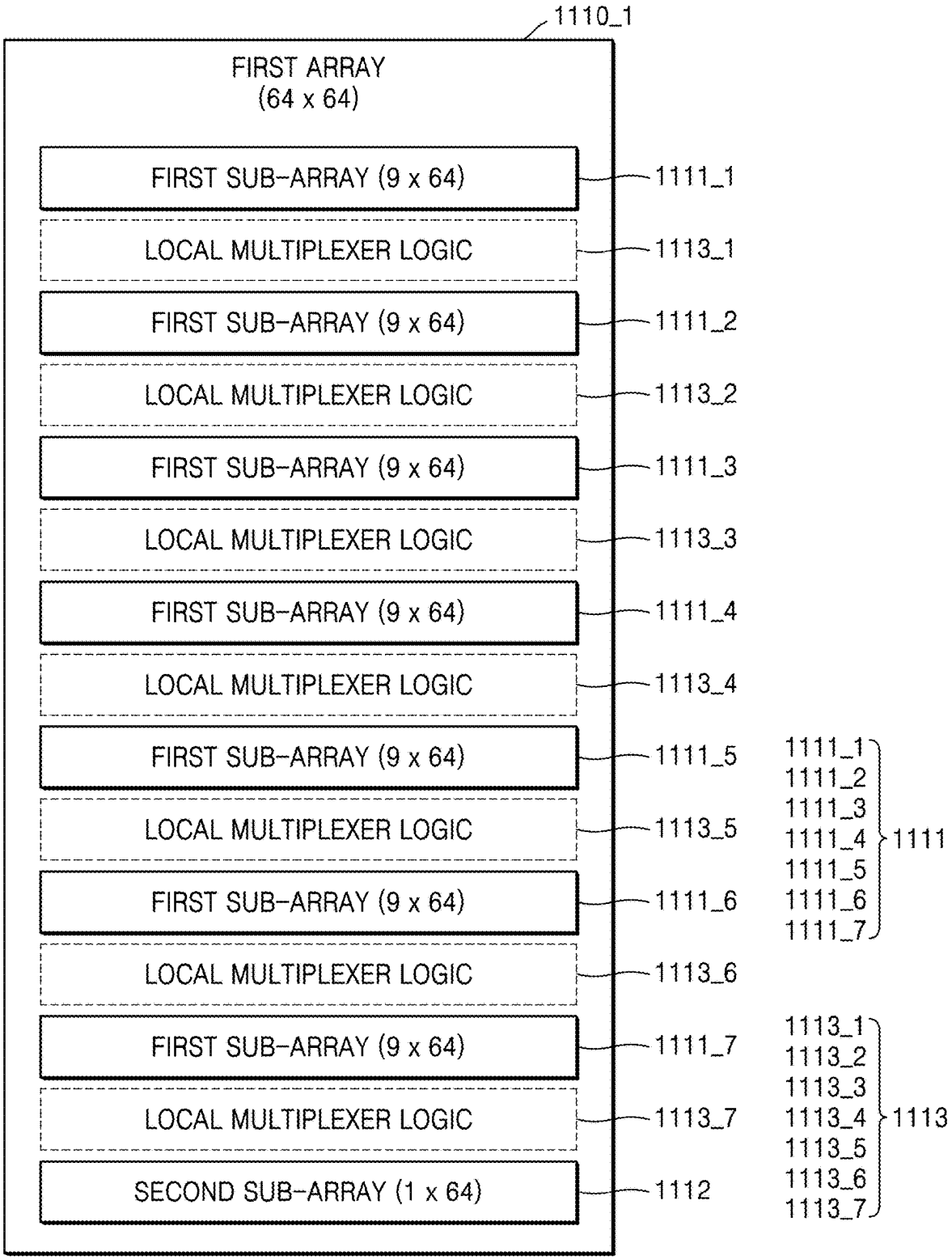

1110_1
FIRST ARRAY
(64 x 64)
FIRST SUB-ARRAY (9 x 64)
1111_1
LOCAL MULTIPLEXER LOGIC
1113_1
FIRST SUB-ARRAY (9 x 64)
1111_2
LOCAL MULTIPLEXER LOGIC
1113_2
FIRST SUB-ARRAY (9 x 64)
1111_3
LOCAL MULTIPLEXER LOGIC
1113_3
FIRST SUB-ARRAY (9 x 64)
1111_4
LOCAL MULTIPLEXER LOGIC
1113_4
FIRST SUB-ARRAY (9 x 64)
1111_5
LOCAL MULTIPLEXER LOGIC
1113_5
FIRST SUB-ARRAY (9 x 64)
1111_6
LOCAL MULTIPLEXER LOGIC
1113_6
FIRST SUB-ARRAY (9 x 64)
1111_7
LOCAL MULTIPLEXER LOGIC
1113_7
SECOND SUB-ARRAY (1 x 64)
1112
1111_1
1111_2
1111_3
1111_4
1111_5
1111_6
1111_7
1111
1113_1
1113_2
1113_3
1113_4
1113_5
1113_6
1113_7
1113

FIG. 4

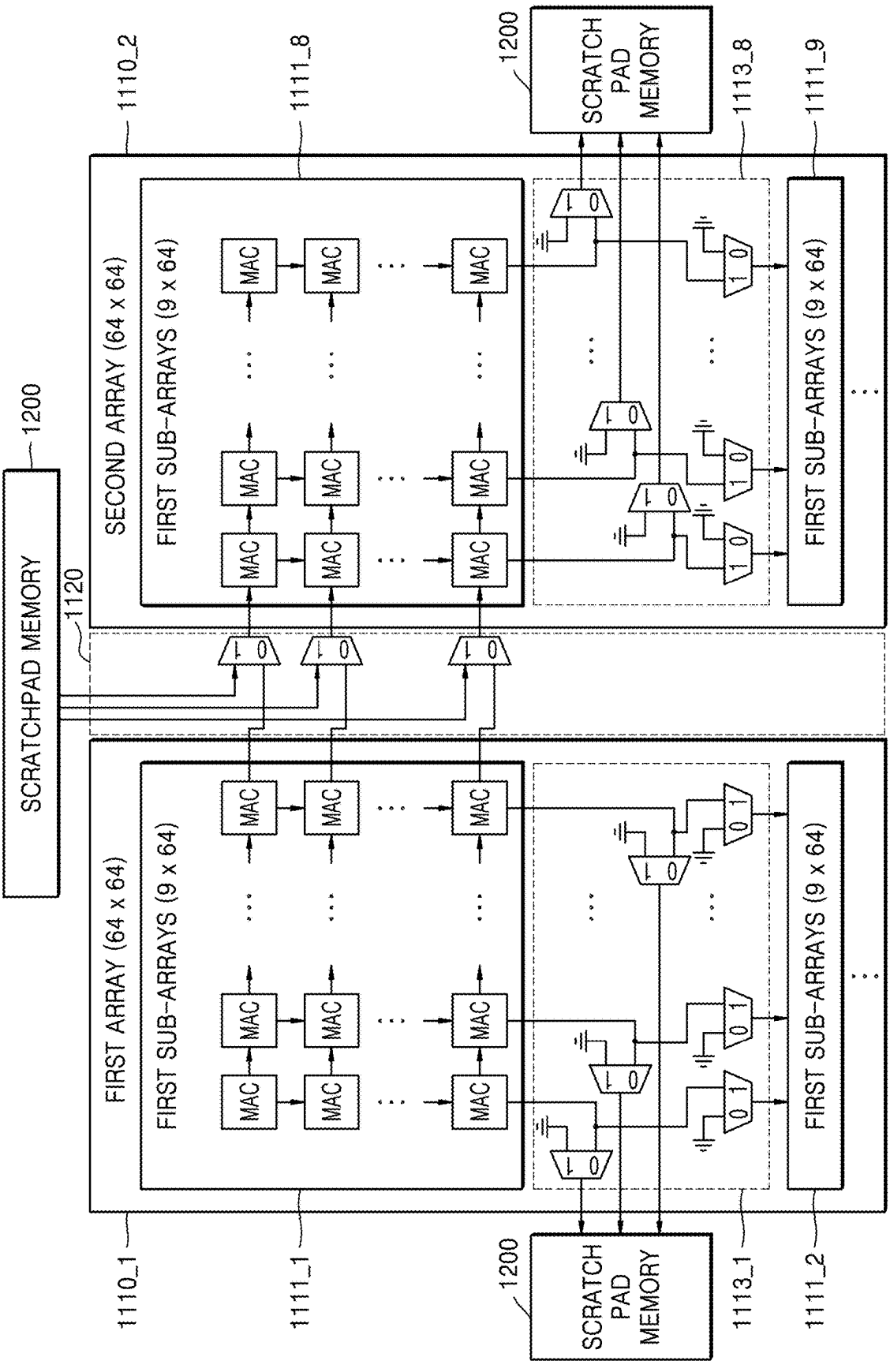

SCRATCHPAD MEMORY
1200
1120
SECOND ARRAY (64 x 64)
FIRST SUB-ARRAYS (9 x 64)
FIRST ARRAY (64 x 64)
FIRST SUB-ARRAYS (9 x 64)
MAC
SCRATCH PAD MEMORY
1200
1110_2
1111_8
1113_8
1111_9
1110_1
1111_1
1113_1
1111_2

FIG. 5

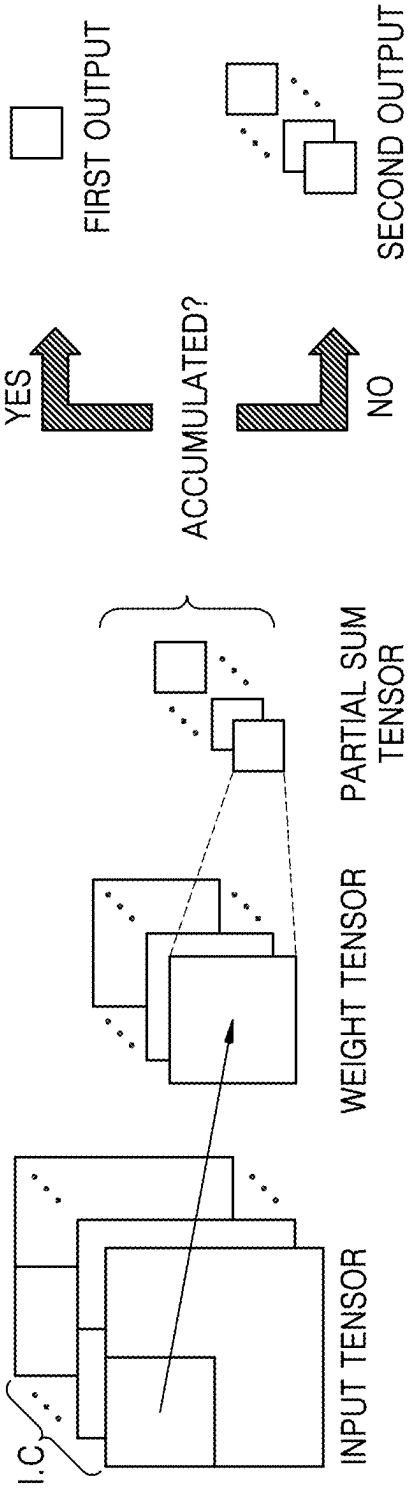

I.C
INPUT TENSOR
WEIGHT TENSOR
PARTIAL SUM TENSOR
ACCUMULATED?
YES
NO
FIRST OUTPUT
SECOND OUTPUT

FIG. 6

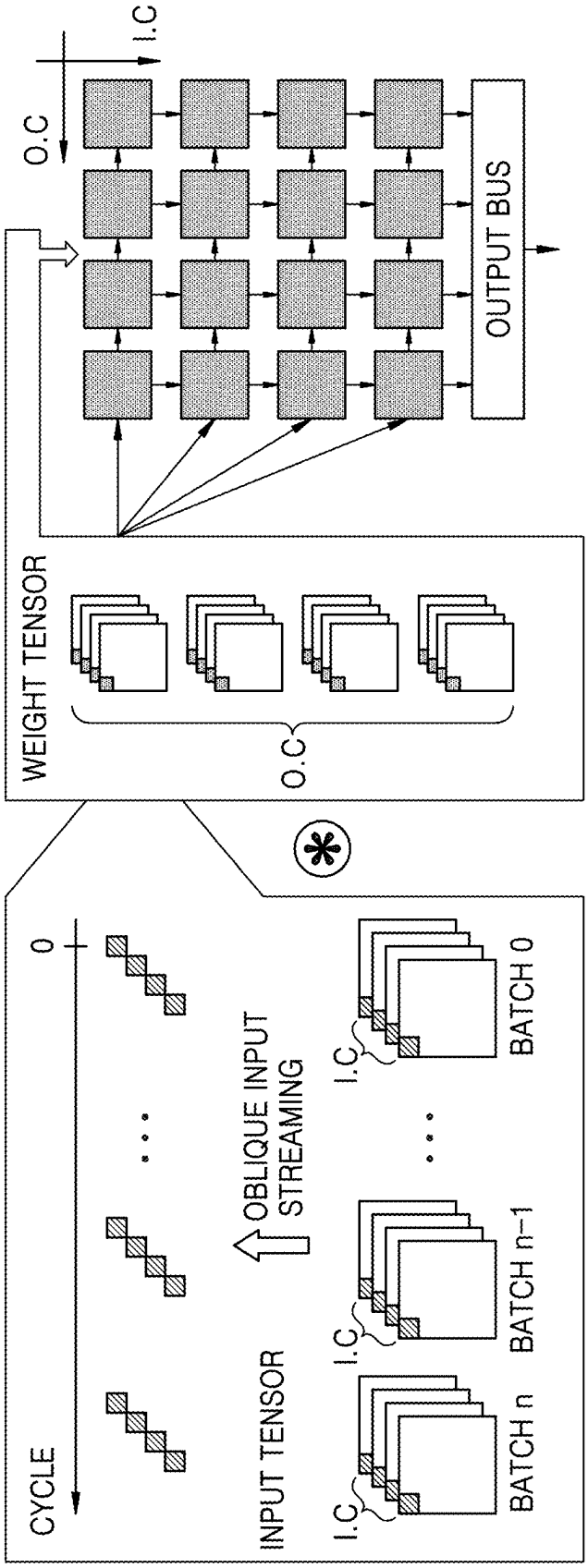

I.C
O.C
OUTPUT BUS
WEIGHT TENSOR
O.C
0
CYCLE
OBLIQUE INPUT STREAMING
INPUT TENSOR
I.C
BATCH n
I.C
BATCH n-1
I.C
BATCH 0

FIG. 7

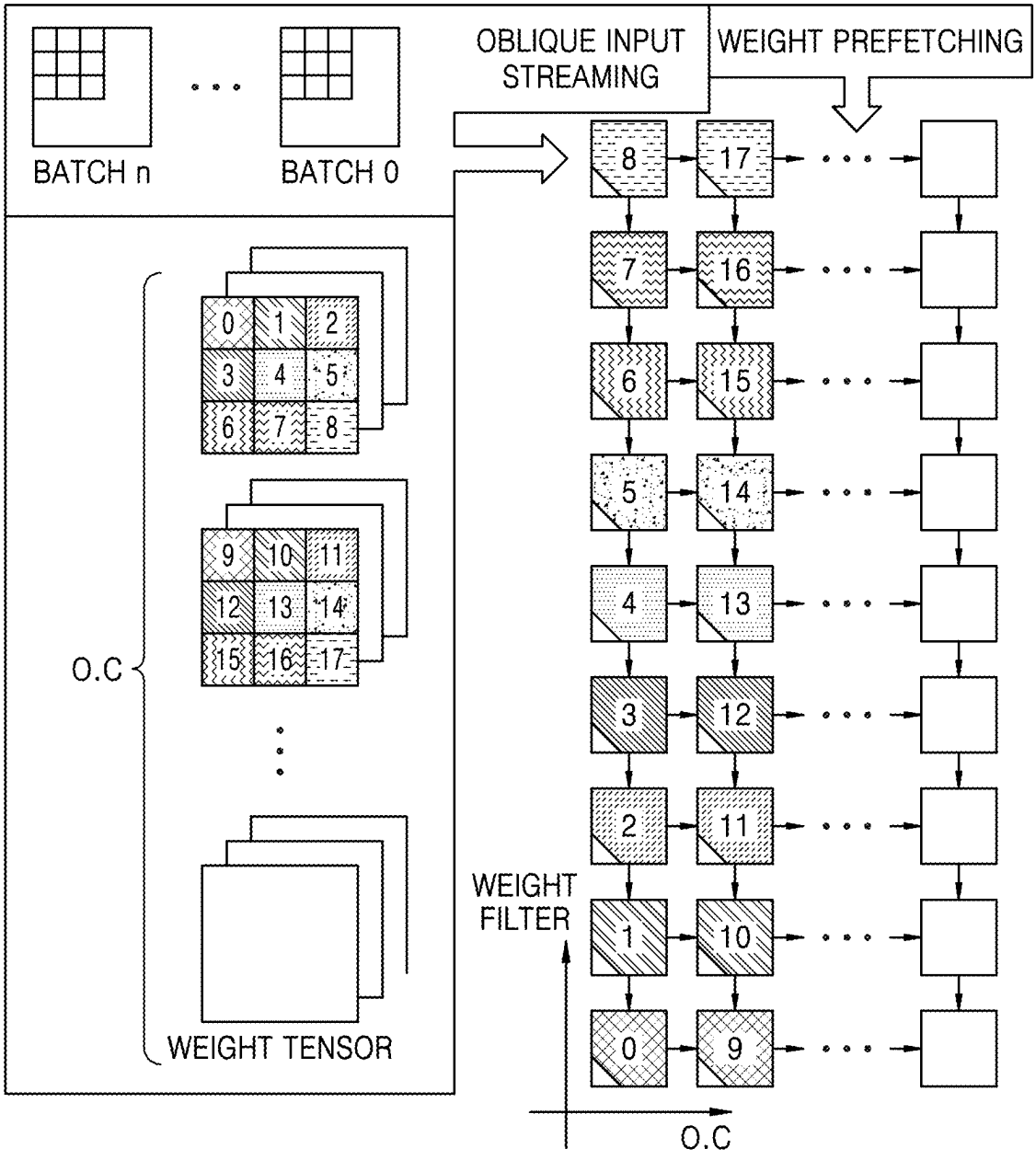

BATCH n
BATCH 0
OBLIQUE INPUT STREAMING
WEIGHT PREFETCHING
O.C
WEIGHT TENSOR
WEIGHT FILTER
O.C

FIG. 8

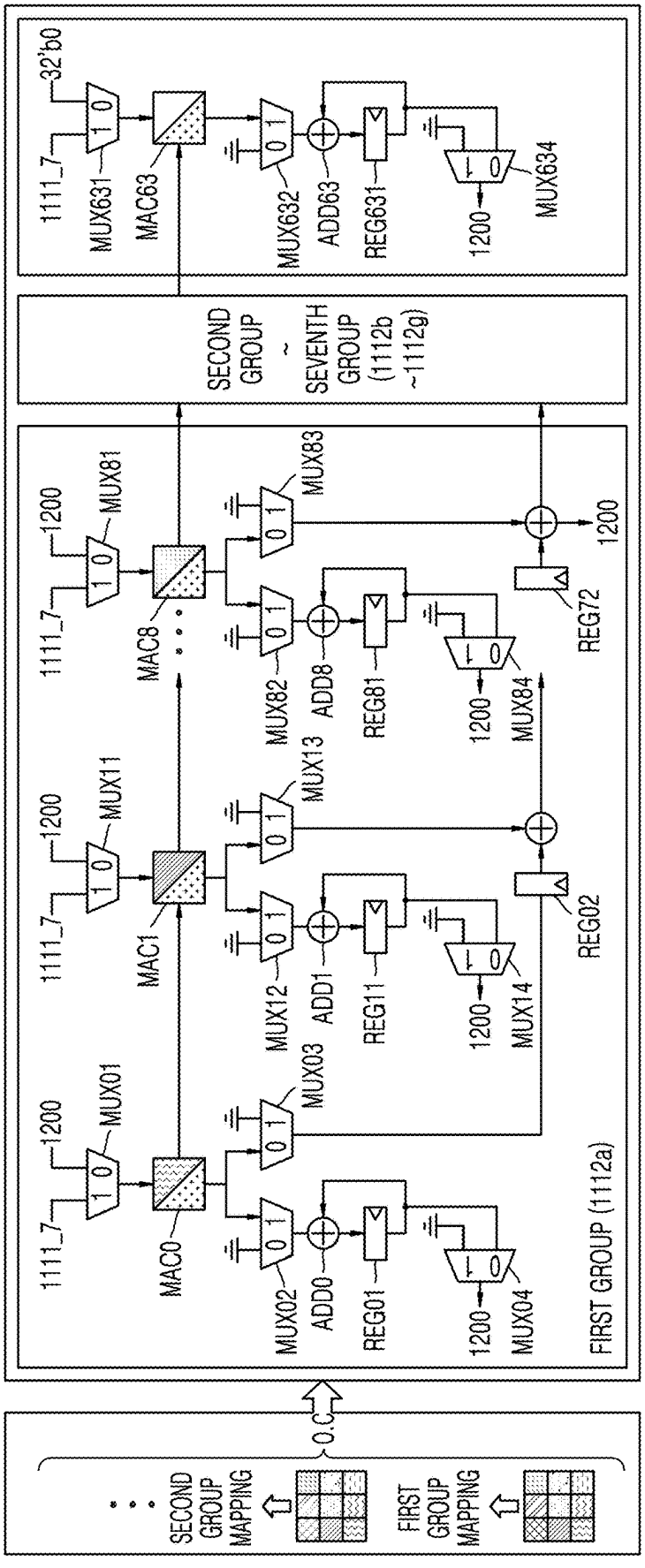

SECOND GROUP MAPPING
FIRST GROUP MAPPING
O.C
FIRST GROUP (1112a)
SECOND GROUP ~ SEVENTH GROUP (1112b ~1112g)
MAC0
MAC1
MAC8
MAC63
MUX01
MUX02
MUX03
MUX04
MUX11
MUX12
MUX13
MUX14
MUX81
MUX82
MUX83
MUX84
MUX631
MUX632
MUX634
ADD0
ADD1
ADD8
ADD63
REG01
REG02
REG11
REG72
REG81
REG631
1111_7
1200
32'b0

FIG. 9

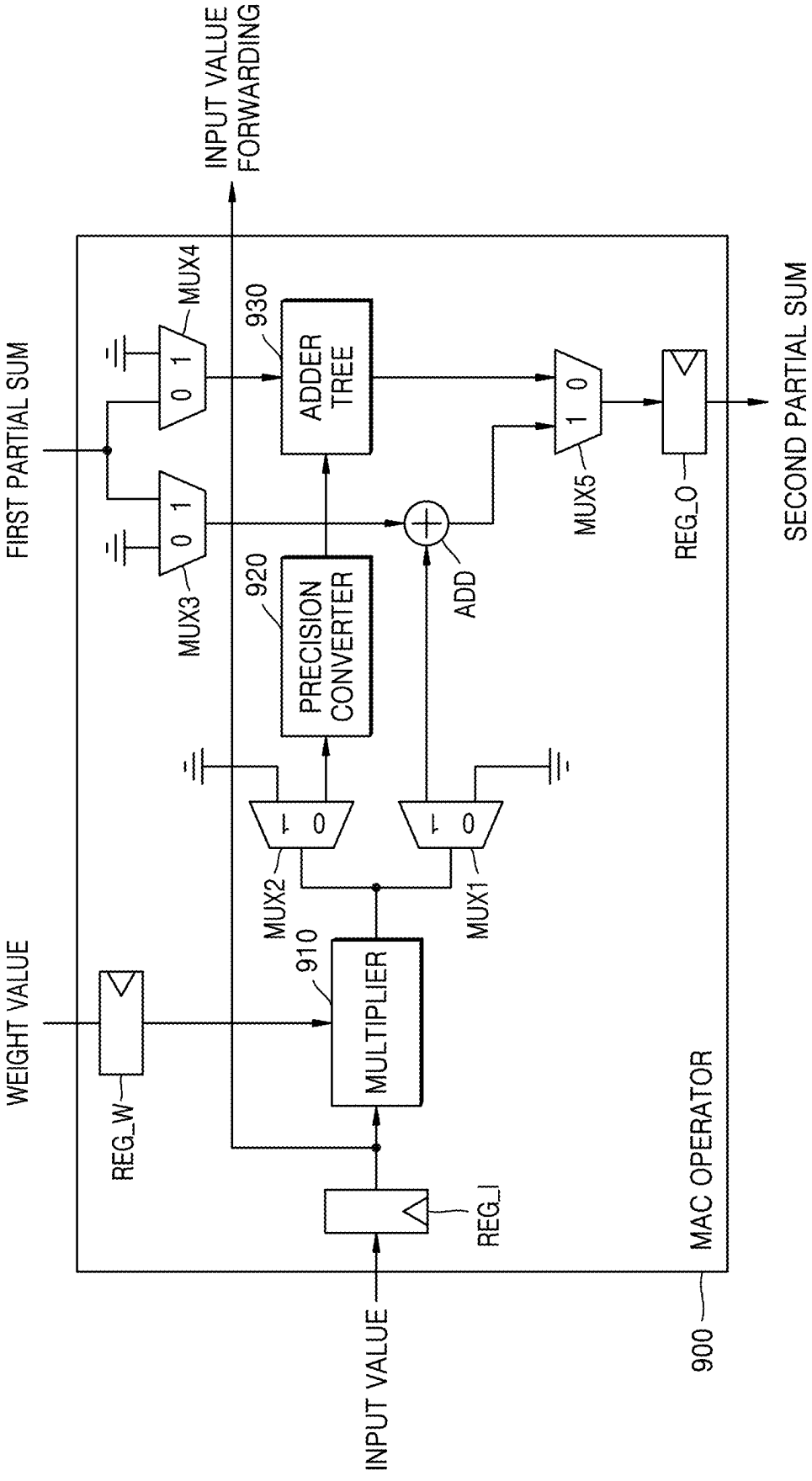

INPUT VALUE FORWARDING
FIRST PARTIAL SUM
WEIGHT VALUE
INPUT VALUE
SECOND PARTIAL SUM
MAC OPERATOR
900
MUX4
930
ADDER TREE
MUX3
920
PRECISION CONVERTER
ADD
MUX5
REG_O
MUX2
MUX1
910
MULTIPLIER
REG_W
REG_I

FIG. 10

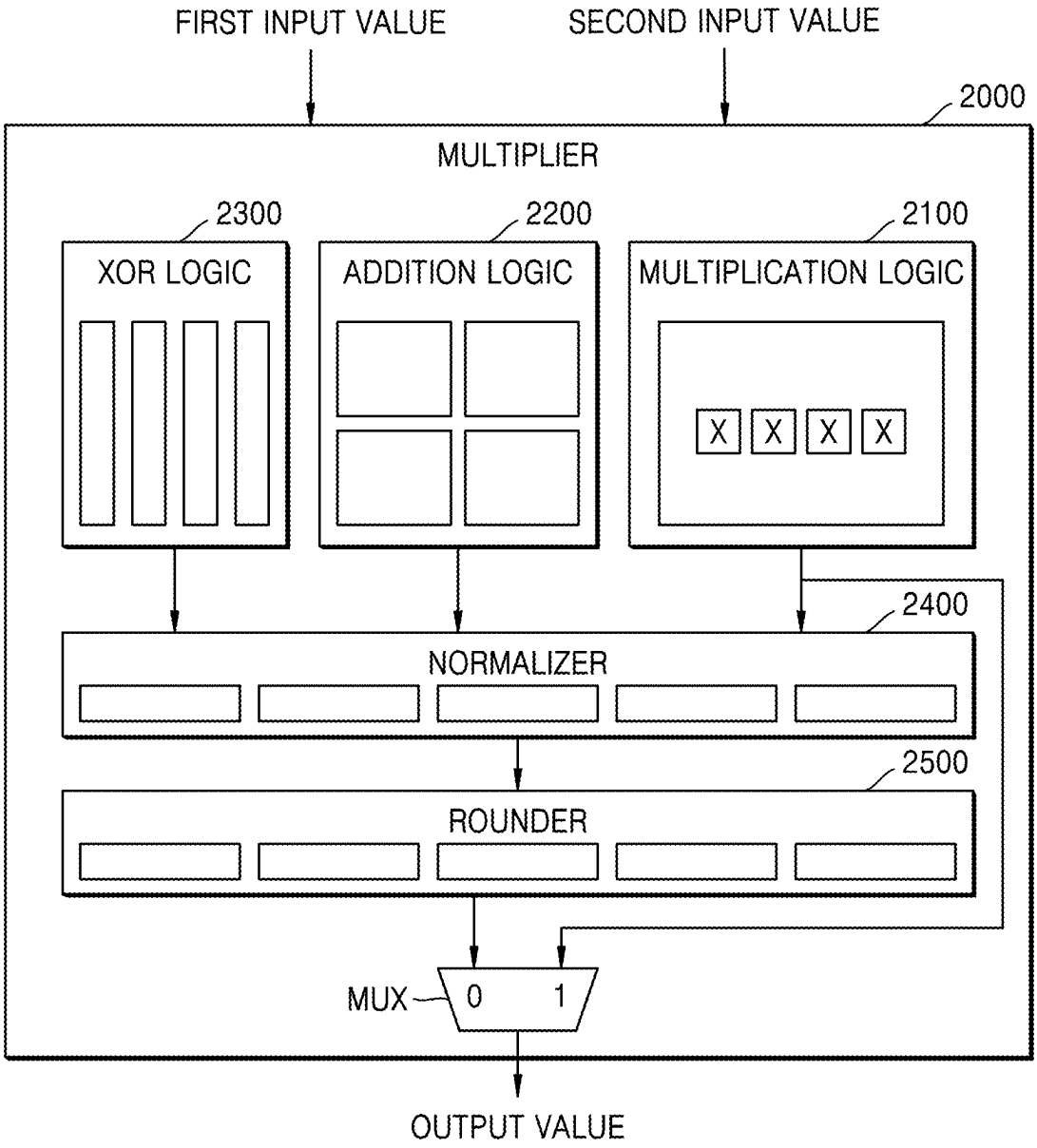

FIRST INPUT VALUE
SECOND INPUT VALUE
2000
MULTIPLIER
2300
XOR LOGIC
2200
ADDITION LOGIC
2100
MULTIPLICATION LOGIC
X
X
X
X
2400
NORMALIZER
2500
ROUNDER
MUX
0
1
OUTPUT VALUE

FIG. 11A

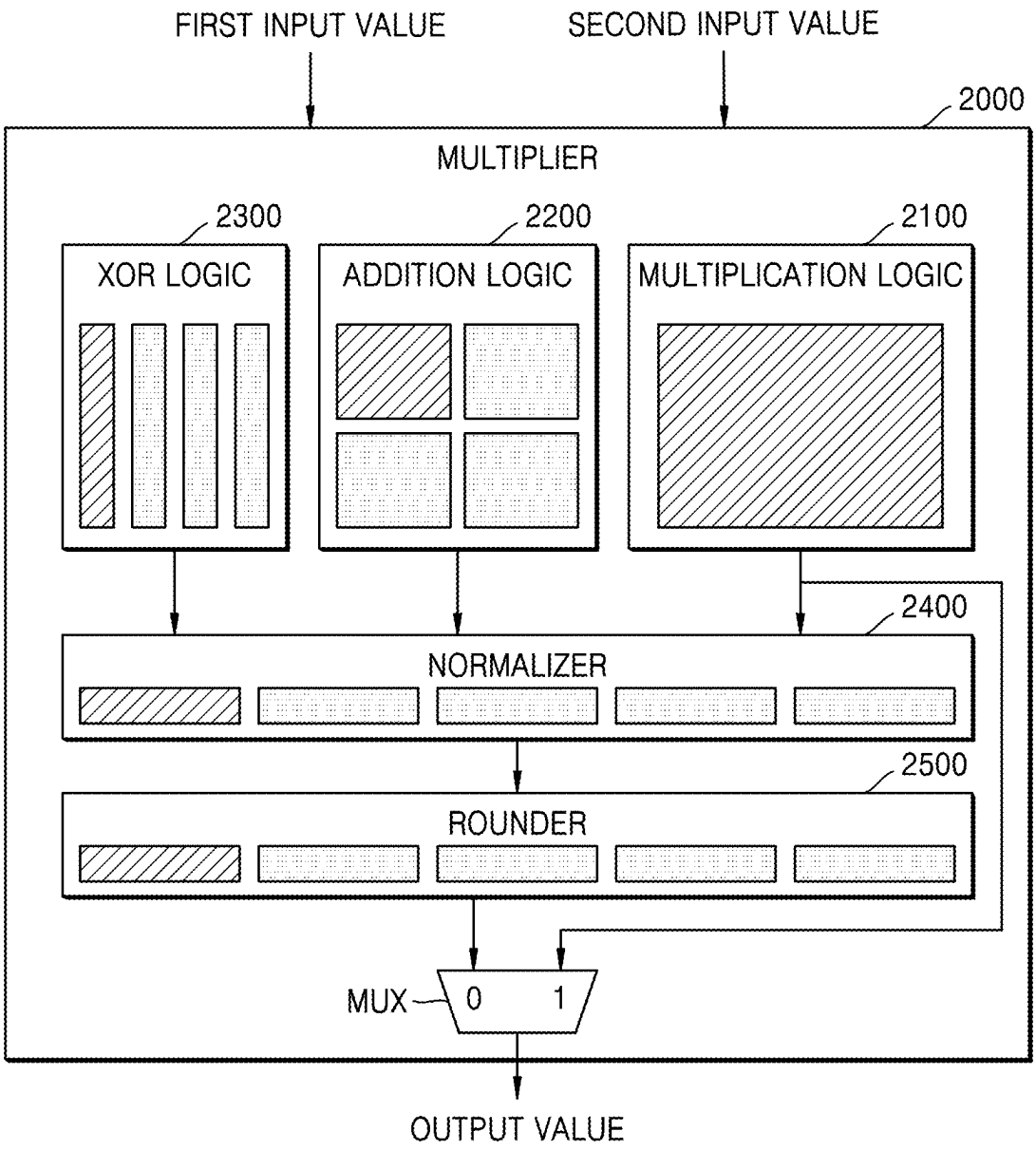

FIRST INPUT VALUE
SECOND INPUT VALUE
2000
MULTIPLIER
2300
XOR LOGIC
2200
ADDITION LOGIC
2100
MULTIPLICATION LOGIC
2400
NORMALIZER
2500
ROUNDER
MUX
0
1
OUTPUT VALUE

FIG. 11B

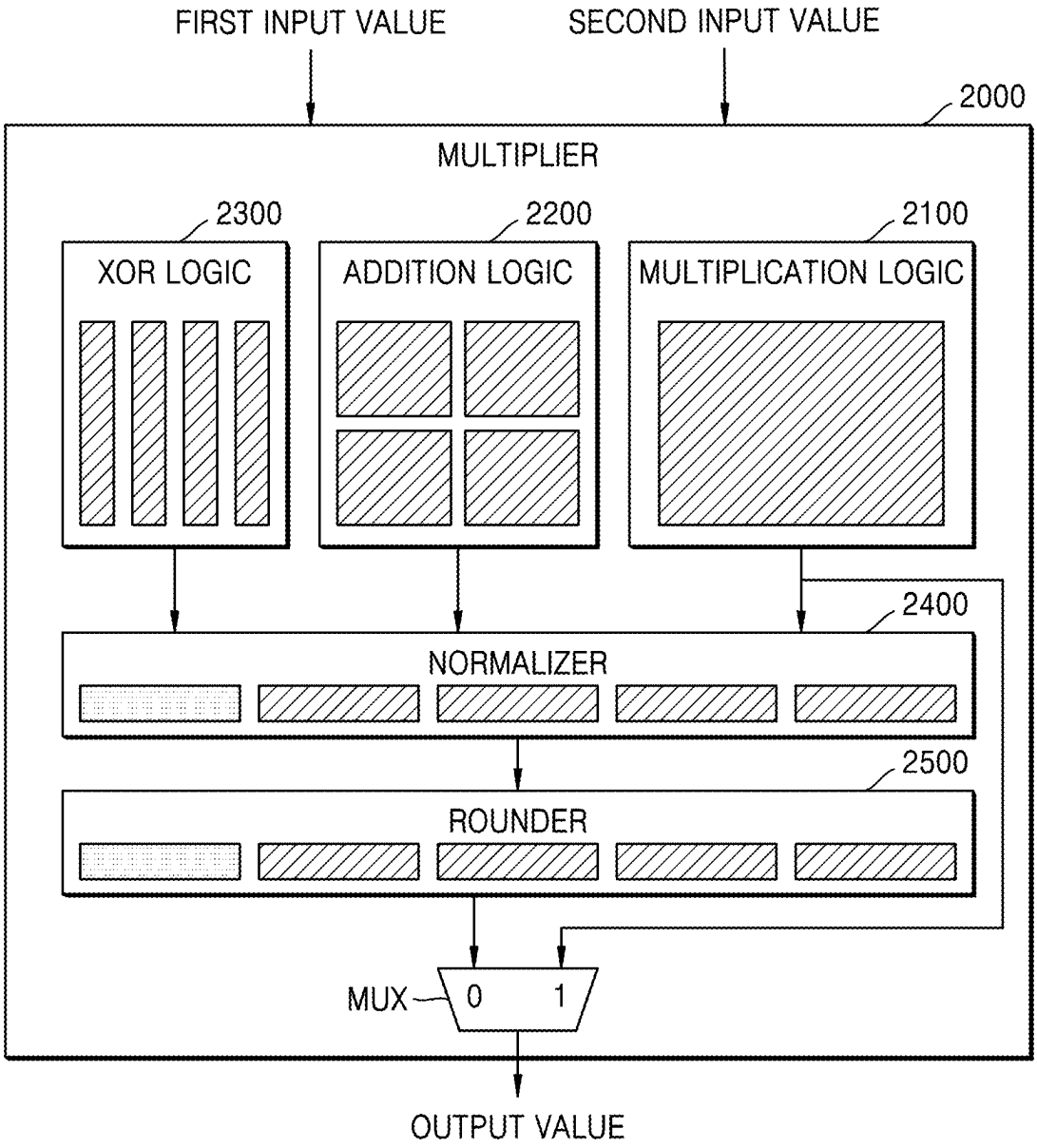

FIRST INPUT VALUE
SECOND INPUT VALUE
2000
MULTIPLIER
2300
XOR LOGIC
2200
ADDITION LOGIC
2100
MULTIPLICATION LOGIC
2400
NORMALIZER
2500
ROUNDER
MUX
0
1
OUTPUT VALUE

FIG. 11C

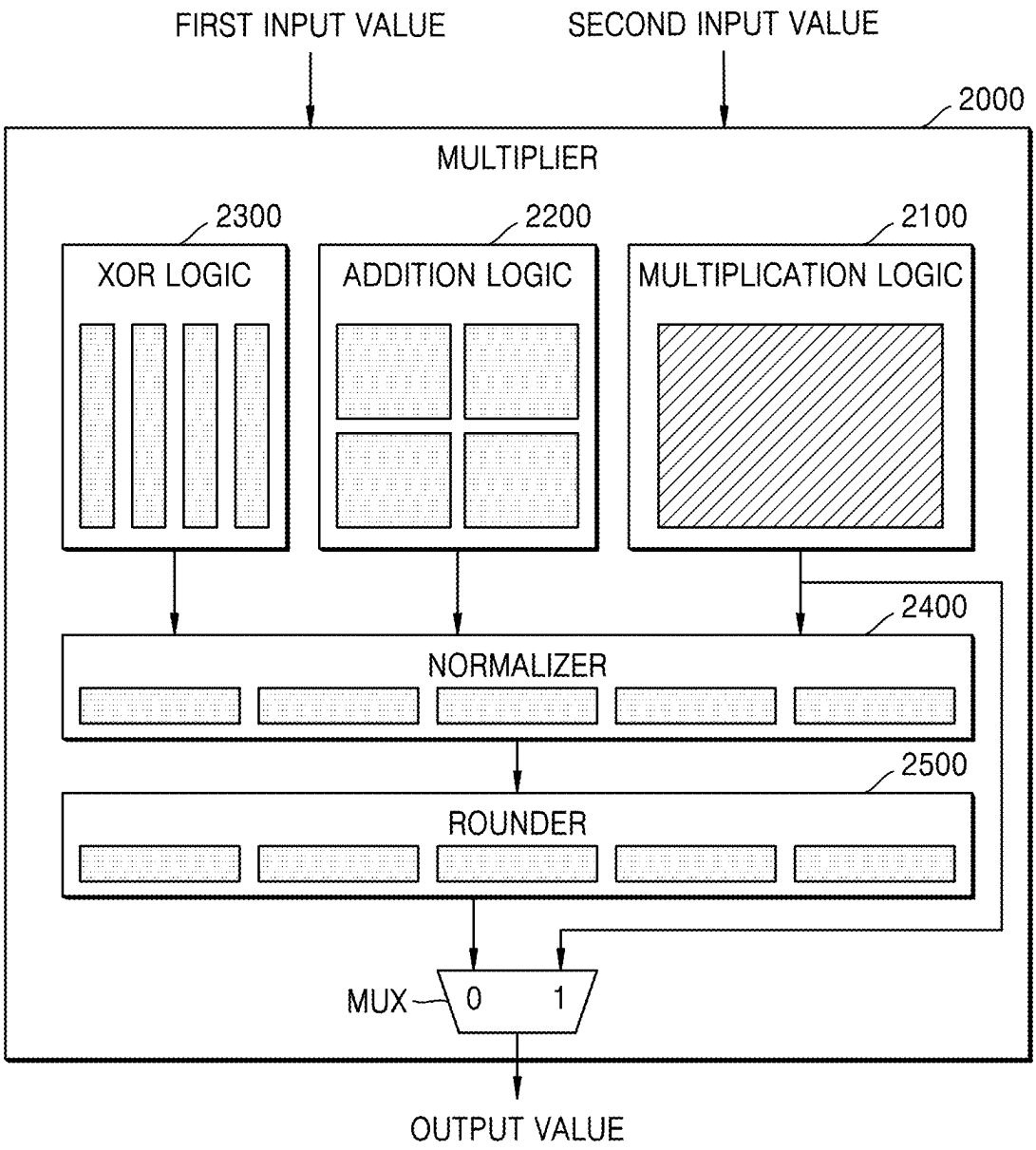

FIRST INPUT VALUE
SECOND INPUT VALUE
2000
MULTIPLIER
2300
XOR LOGIC
2200
ADDITION LOGIC
2100
MULTIPLICATION LOGIC
2400
NORMALIZER
2500
ROUNDER
MUX
0
1
OUTPUT VALUE

FIG. 12

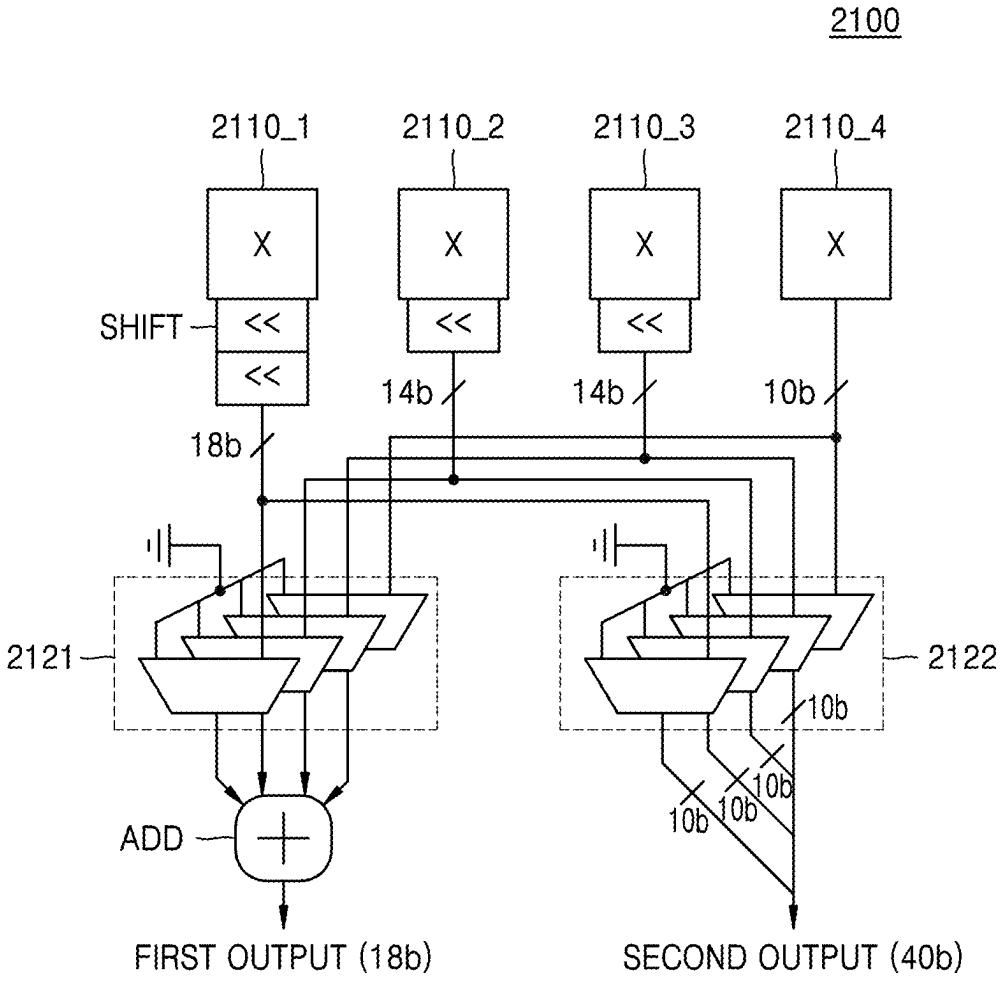

2100
2110_1
2110_2
2110_3
2110_4
X
X
X
X
SHIFT
<<
<<
<<
<<
14b
14b
10b
18b
2121
2122
10b
10b
10b
10b
ADD
FIRST OUTPUT (18b)
SECOND OUTPUT (40b)

FIG. 13

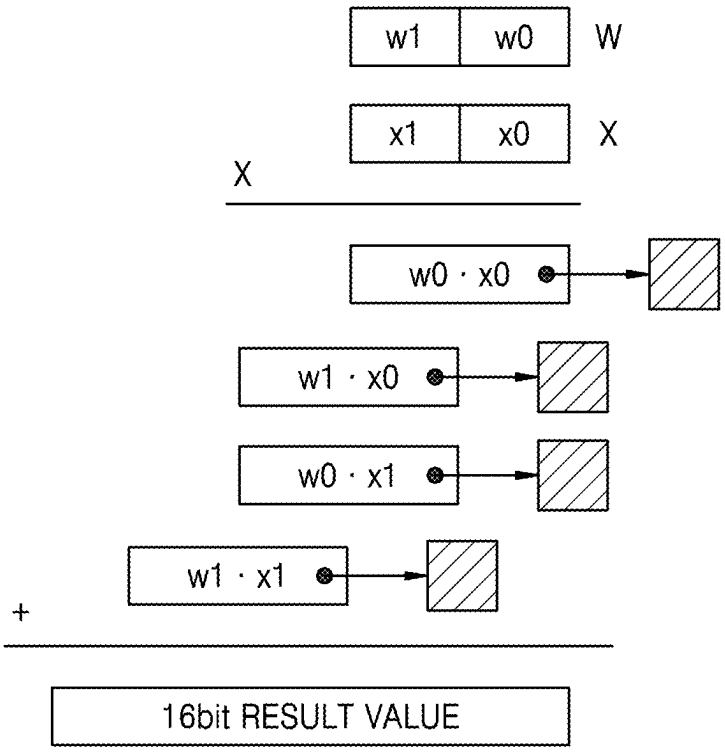


FIG. 14A

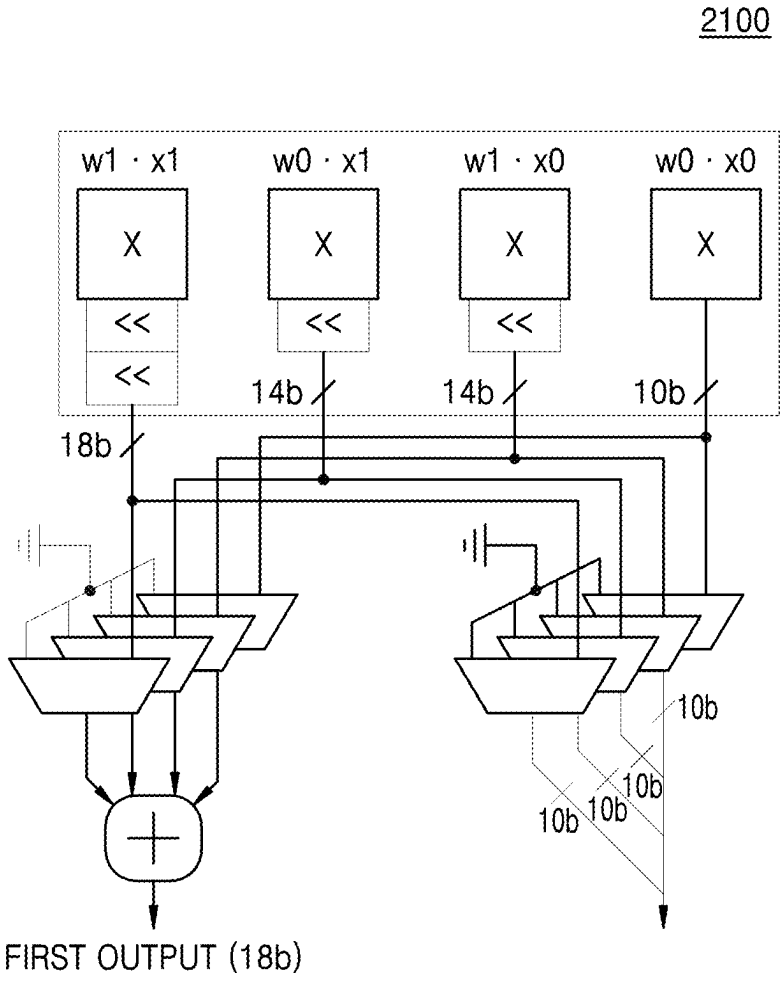

2100
w1 · x1
w0 · x1
w1 · x0
w0 · x0
X
X
X
X
<<
<<
<<
<<
14b
14b
10b
18b
10b
10b
10b
10b
FIRST OUTPUT (18b)

FIG. 14B

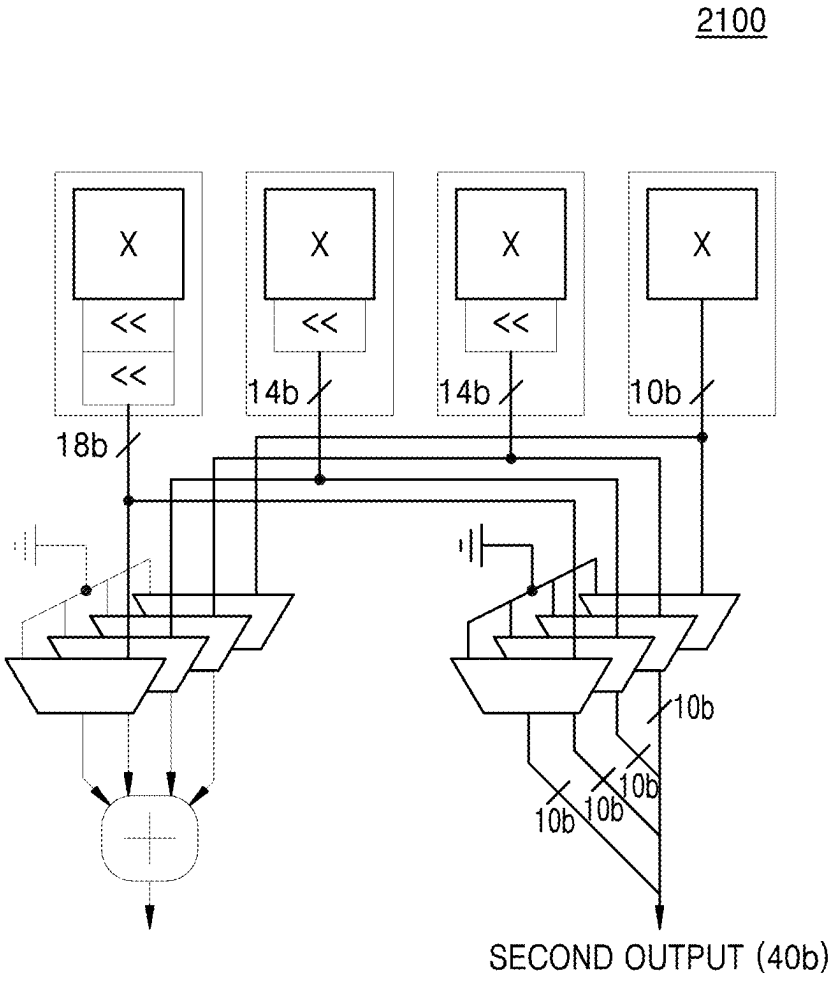


FIG. 15

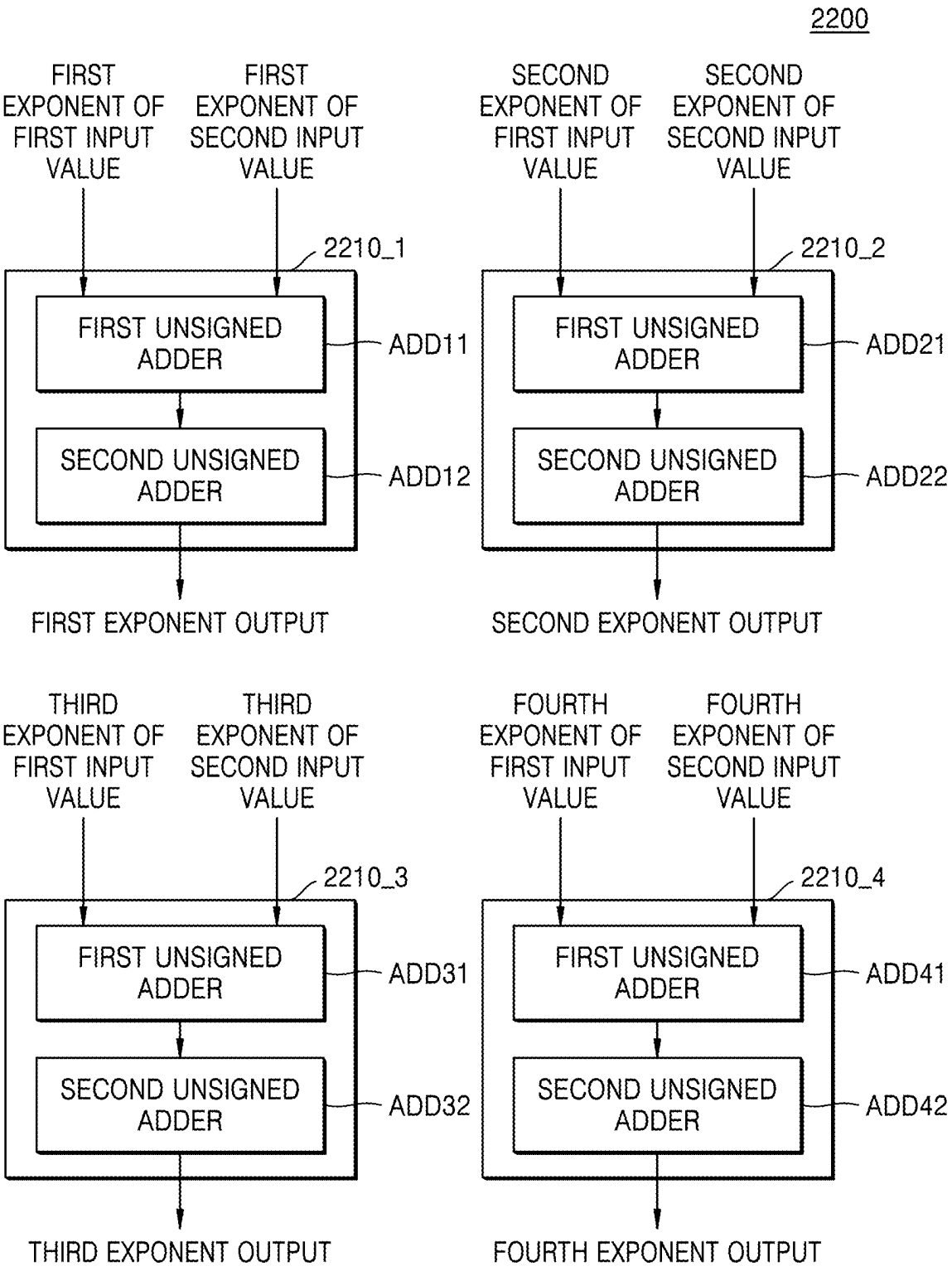

2200
FIRST EXPONENT OF FIRST INPUT VALUE
FIRST EXPONENT OF SECOND INPUT VALUE
2210_1
FIRST UNSIGNED ADDER
ADD11
SECOND UNSIGNED ADDER
ADD12
FIRST EXPONENT OUTPUT
SECOND EXPONENT OF FIRST INPUT VALUE
SECOND EXPONENT OF SECOND INPUT VALUE
2210_2
FIRST UNSIGNED ADDER
ADD21
SECOND UNSIGNED ADDER
ADD22
SECOND EXPONENT OUTPUT
THIRD EXPONENT OF FIRST INPUT VALUE
THIRD EXPONENT OF SECOND INPUT VALUE
2210_3
FIRST UNSIGNED ADDER
ADD31
SECOND UNSIGNED ADDER
ADD32
THIRD EXPONENT OUTPUT
FOURTH EXPONENT OF FIRST INPUT VALUE
FOURTH EXPONENT OF SECOND INPUT VALUE
2210_4
FIRST UNSIGNED ADDER
ADD41
SECOND UNSIGNED ADDER
ADD42
FOURTH EXPONENT OUTPUT

FIG. 16

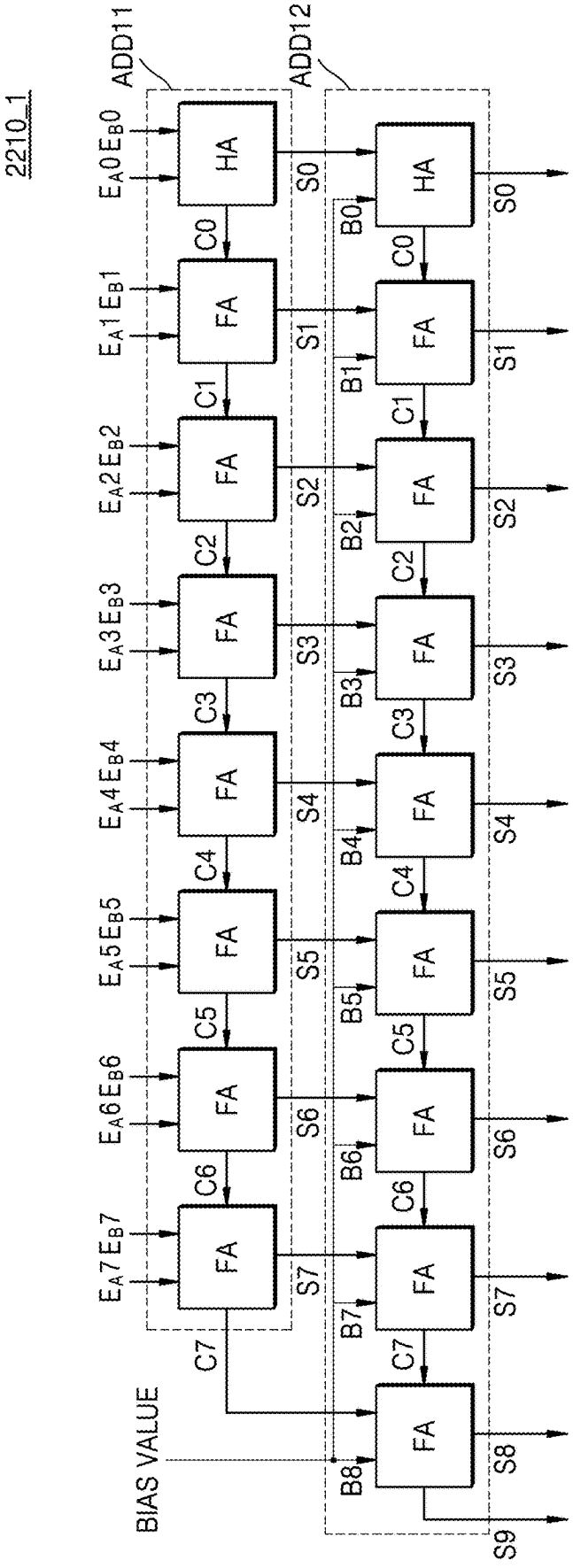


FIG. 17

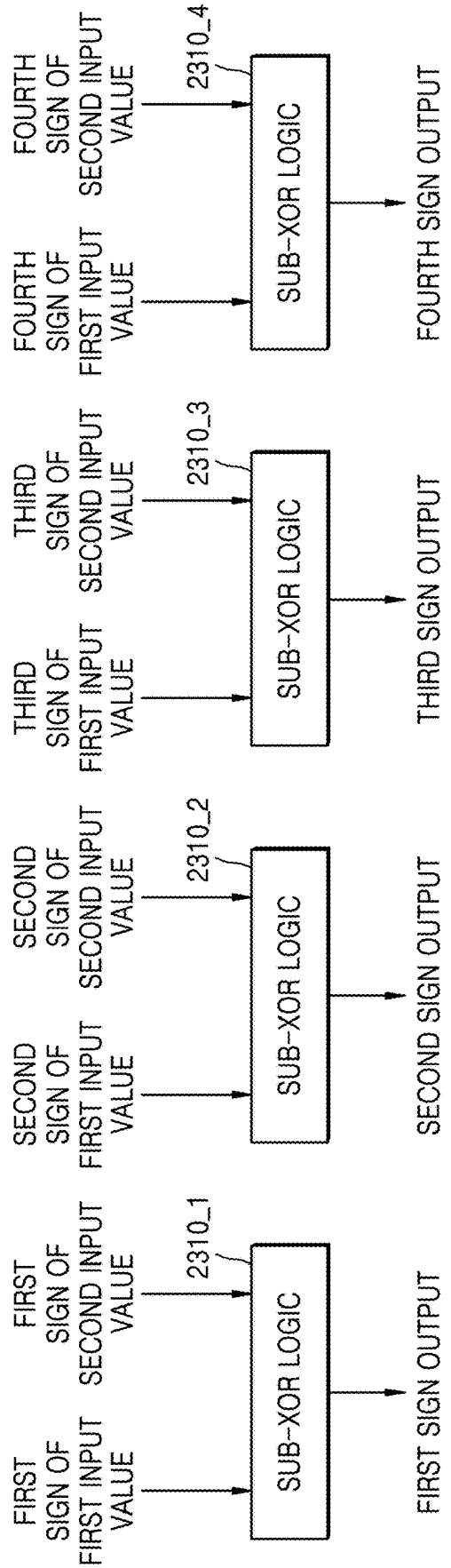

FIRST SIGN OF FIRST INPUT VALUE
FIRST SIGN OF SECOND INPUT VALUE
2310_1
SUB-XOR LOGIC
FIRST SIGN OUTPUT
SECOND SIGN OF FIRST INPUT VALUE
SECOND SIGN OF SECOND INPUT VALUE
2310_2
SUB-XOR LOGIC
SECOND SIGN OUTPUT
THIRD SIGN OF FIRST INPUT VALUE
THIRD SIGN OF SECOND INPUT VALUE
2310_3
SUB-XOR LOGIC
THIRD SIGN OUTPUT
FOURTH SIGN OF FIRST INPUT VALUE
FOURTH SIGN OF SECOND INPUT VALUE
2310_4
SUB-XOR LOGIC
FOURTH SIGN OUTPUT

FIG. 18

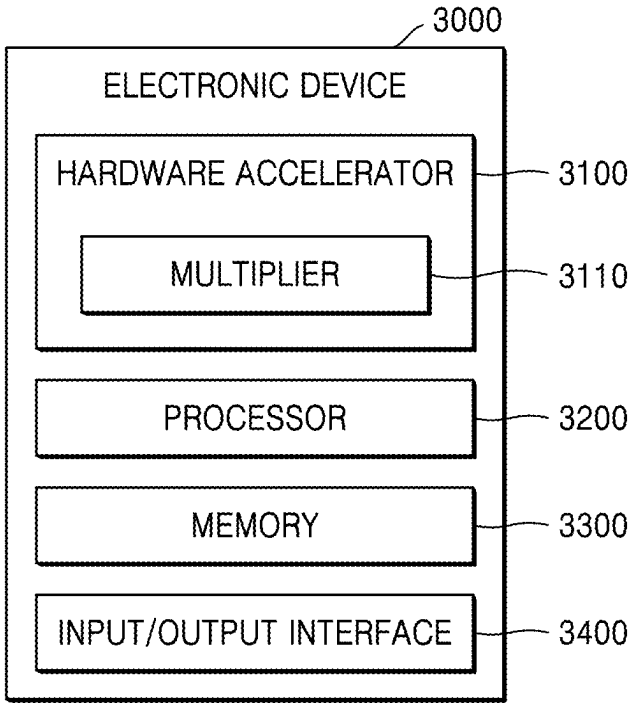

3000
ELECTRONIC DEVICE
HARDWARE ACCELERATOR
3100
MULTIPLIER
3110
PROCESSOR
3200
MEMORY
3300
INPUT/OUTPUT INTERFACE
3400

FIG. 19

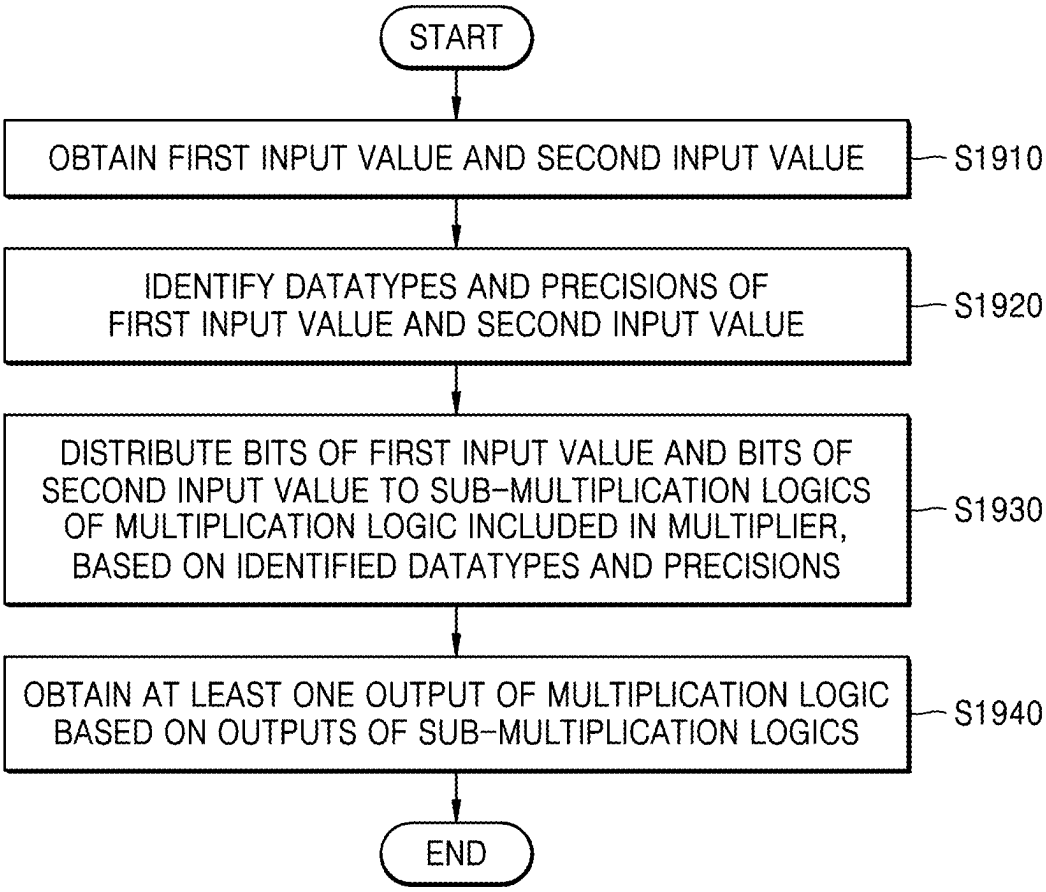

START
OBTAIN FIRST INPUT VALUE AND SECOND INPUT VALUE
S1910
IDENTIFY DATATYPES AND PRECISIONS OF FIRST INPUT VALUE AND SECOND INPUT VALUE
S1920
DISTRIBUTE BITS OF FIRST INPUT VALUE AND BITS OF SECOND INPUT VALUE TO SUB-MULTIPLICATION LOGICS OF MULTIPLICATION LOGIC INCLUDED IN MULTIPLIER, BASED ON IDENTIFIED DATATYPES AND PRECISIONS
S1930
OBTAIN AT LEAST ONE OUTPUT OF MULTIPLICATION LOGIC BASED ON OUTPUTS OF SUB-MULTIPLICATION LOGICS
S1940
END

FIG. 20

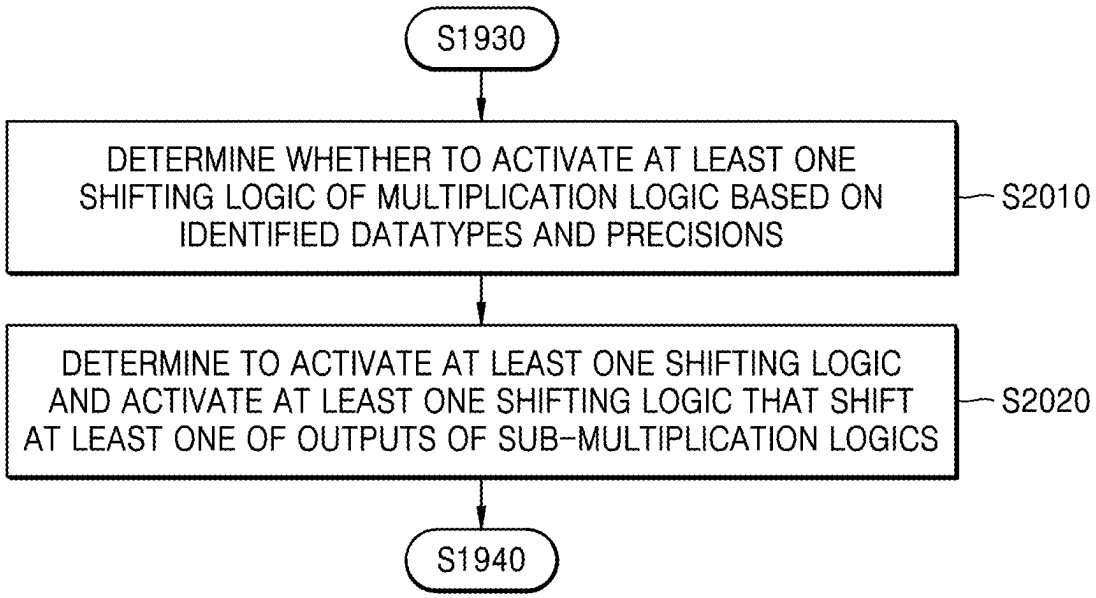

S1930
DETERMINE WHETHER TO ACTIVATE AT LEAST ONE SHIFTING LOGIC OF MULTIPLICATION LOGIC BASED ON IDENTIFIED DATATYPES AND PRECISIONS
S2010
DETERMINE TO ACTIVATE AT LEAST ONE SHIFTING LOGIC AND ACTIVATE AT LEAST ONE SHIFTING LOGIC THAT SHIFT AT LEAST ONE OF OUTPUTS OF SUB-MULTIPLICATION LOGICS
S2020
S1940

FIG. 21

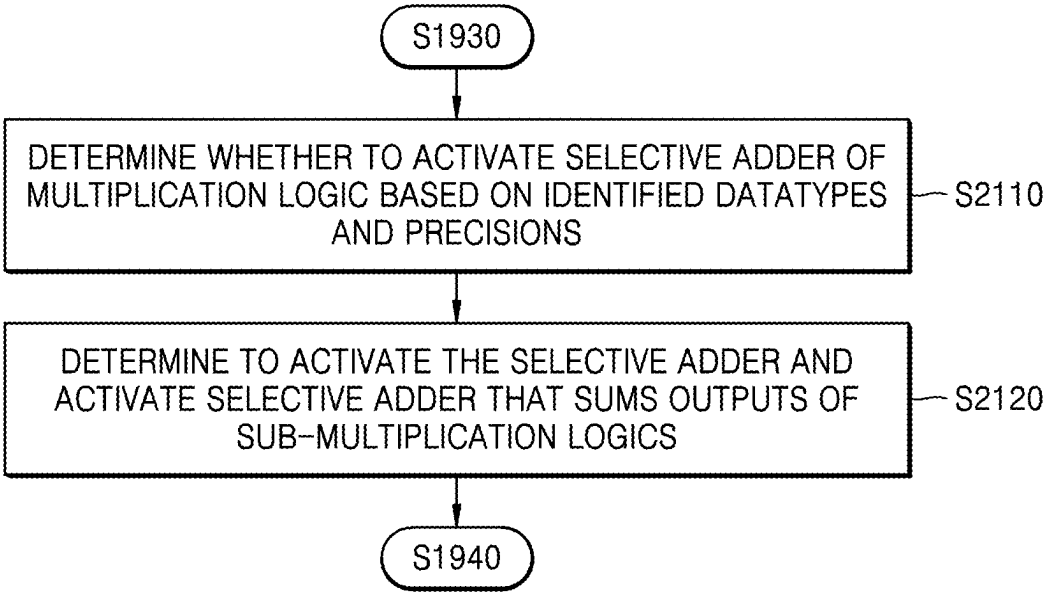

S1930
DETERMINE WHETHER TO ACTIVATE SELECTIVE ADDER OF MULTIPLICATION LOGIC BASED ON IDENTIFIED DATATYPES AND PRECISIONS
S2110
DETERMINE TO ACTIVATE THE SELECTIVE ADDER AND ACTIVATE SELECTIVE ADDER THAT SUMS OUTPUTS OF SUB-MULTIPLICATION LOGICS
S2120
S1940

FIG. 22

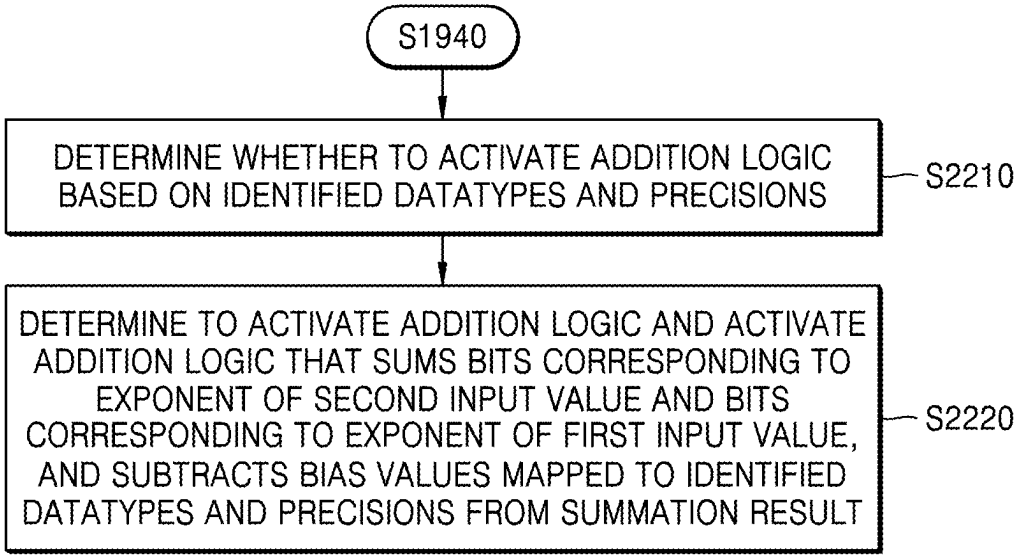

S1940
DETERMINE WHETHER TO ACTIVATE ADDITION LOGIC BASED ON IDENTIFIED DATATYPES AND PRECISIONS
S2210
DETERMINE TO ACTIVATE ADDITION LOGIC AND ACTIVATE ADDITION LOGIC THAT SUMS BITS CORRESPONDING TO EXPONENT OF SECOND INPUT VALUE AND BITS CORRESPONDING TO EXPONENT OF FIRST INPUT VALUE, AND SUBTRACTS BIAS VALUES MAPPED TO IDENTIFIED DATATYPES AND PRECISIONS FROM SUMMATION RESULT
S2220

FIG. 23

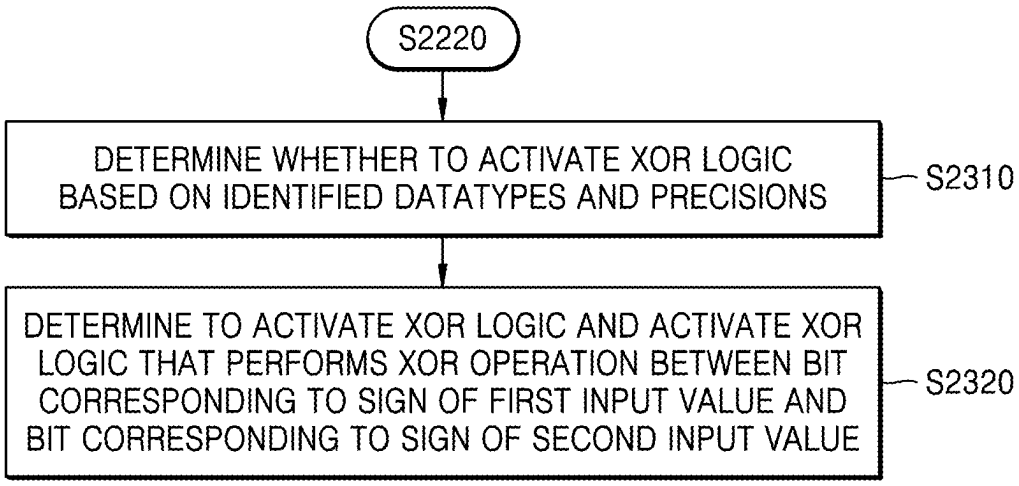
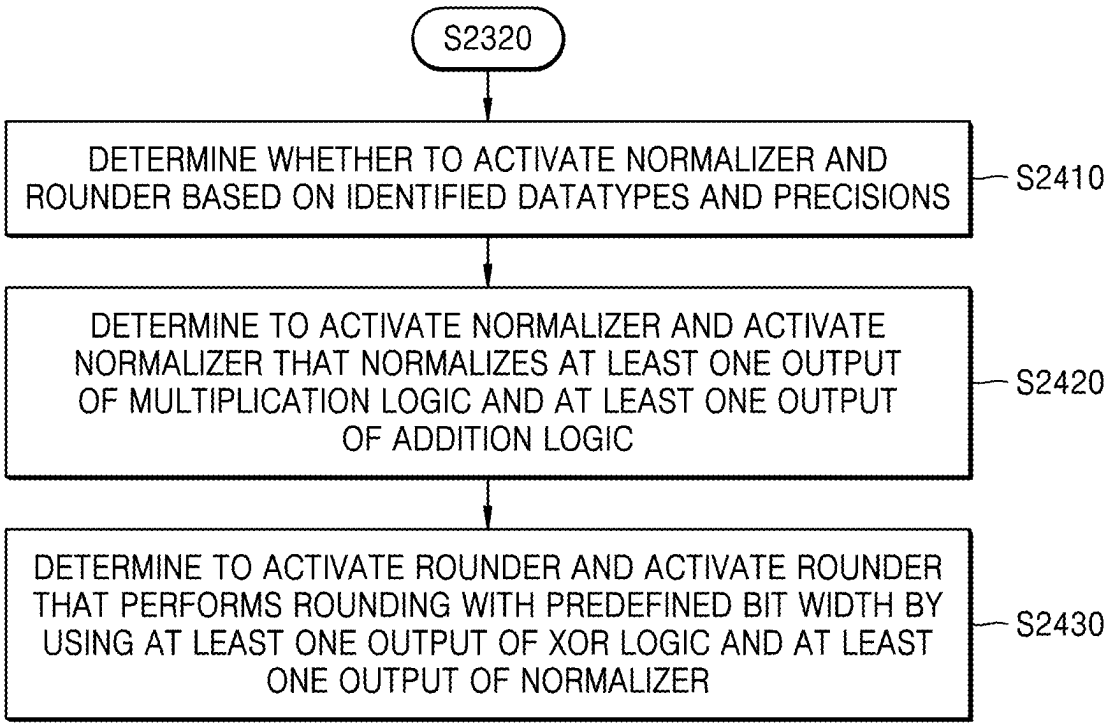

S2220
DETERMINE WHETHER TO ACTIVATE XOR LOGIC BASED ON IDENTIFIED DATATYPES AND PRECISIONS
S2310
DETERMINE TO ACTIVATE XOR LOGIC AND ACTIVATE XOR LOGIC THAT PERFORMS XOR OPERATION BETWEEN BIT CORRESPONDING TO SIGN OF FIRST INPUT VALUE AND BIT CORRESPONDING TO SIGN OF SECOND INPUT VALUE
S2320

FIG. 24


S2320
DETERMINE WHETHER TO ACTIVATE NORMALIZER AND ROUNDER BASED ON IDENTIFIED DATATYPES AND PRECISIONS
S2410
DETERMINE TO ACTIVATE NORMALIZER AND ACTIVATE NORMALIZER THAT NORMALIZES AT LEAST ONE OUTPUT OF MULTIPLICATION LOGIC AND AT LEAST ONE OUTPUT OF ADDITION LOGIC
S2420
DETERMINE TO ACTIVATE ROUNDER AND ACTIVATE ROUNDER THAT PERFORMS ROUNDING WITH PREDEFINED BIT WIDTH BY USING AT LEAST ONE OUTPUT OF XOR LOGIC AND AT LEAST ONE OUTPUT OF NORMALIZER
S2430

MULTIPLIER SUPPORTING VARIOUS PRECISIONS AND DATATYPES AND OPERATING METHOD THEREOF

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is based on and claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2024-0031558, filed on Mar. 5, 2024, in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

1. Field

[0002] The disclosure relates to a multiply-accumulate (MAC) array and a hardware accelerator including the same, and more particularly, to a method of mapping data to a MAC operator that performs operations of a deep neural network model, an arrangement structure of MAC operators being included in a MAC array, and a method of transferring and receiving data between the MAC operators.

[0003] The disclosure relates to a multiplier that supports various precisions and datatypes and an operation method thereof, and more particularly, to a multiplier capable of identifying precisions and datatypes of values input in bit units and adaptively performing a multiplication operation and an operation method of the multiplier.

[0004] This research has been conducted with support from the Samsung Future Technology Promotion Project (Project Number: SRFC-IT1902-03).

2. Description of the Related Art

[0005] High-performance computing systems and continuously growing open-source datasets have led to extremely quick advances of artificial intelligence technology. In addition, along with the improved accuracy in artificial intelligence technology, artificial intelligence technology is used in many applications, such as computer vision, language modeling, and autonomous driving.

[0006] To use artificial intelligence for applications, training processes are required. In artificial intelligence technology, training refers to a process of updating weights of artificial intelligence models (for example, deep neural networks (DNNs)) by using specific datasets. As the weights are better updated, artificial intelligence models may perform given tasks better.

[0007] However, because training processes require extremely large amounts of calculation, training via central processing units (CPUs) takes extremely much time. Although graphics processing units (GPUs) facilitate parallel processing and thus less training time is required than in the case of CPUs, GPUs exhibit low usage due to the structural nature thereof. Recently, to overcome the drawbacks of CPUs and GPUs, a lot of dedicated hardware accelerators for performing calculations in DNNs have been proposed.

[0008] In addition, as the types of DNN models become more diverse and multiple users request operations of DNN models in a multitenant environment, the demand for parallel processing functions of dedicated hardware accelerators is increasing.

[0009] Also, as datatypes and precisions of values required for operations of DNN models become more diverse, MAC operators support various datatypes and precisions, and thus, the demand for area-efficient hardware accelerators is increasing.

SUMMARY

[0010] Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments of the disclosure.

[0011] According to an aspect of the disclosure, an electronic device includes a multiplier including a multiplication logic, a memory storing at least one instruction, and at least one processor configured to execute the at least one instruction and to obtain a first input value and a second input value, identify datatypes and precisions of the first input value and the second input value, based on the identified datatypes and precisions, distribute bits of the first input value and bits of the second input value to sub-multiplication logics of the multiplication logic, and obtain at least one output of the multiplication logic based on outputs of the sub-multiplication logics.

[0012] According to another aspect of the disclosure, an operating method of a multiplier includes obtaining a first input value and a second input value, identifying datatypes and precisions of the first input value and the second input value, based on the identified datatypes and precisions, distributing bits of the first input value and bits of the second input value to sub-multiplication logics of a multiplication logic included in the multiplier, and obtaining at least one output of the multiplication logic based on outputs of the sub-multiplication logics.

[0013] According to another aspect of the disclosure, a multiplier includes a multiplication logic configured to, based on datatypes of a first input value and a second input value being integer types, perform a multiplication between the first input value and the second input value, and based on the datatypes being floating-point types, perform a multiplication between a mantissa of the first input value and a mantissa of the second input value, an addition logic configured to, based on the datatypes being floating-point types, sum an exponent of the first input value and an exponent of the second input value, and subtract bias values mapped to bit widths of the exponent of the first input value and the exponent of the second input value from a summation result, an XOR logic configured to, based on the datatypes being floating-point types, perform an XOR operation between a sign of the first input value and a sign of the second input value, a normalizer configured to, based on the datatypes being floating-point types, perform normalization an output of the multiplication logic and an output of the addition logic, a rounder configured to, based on the datatypes being floating-point types, perform rounding with a predefined bit width by using an output of the normalizer and an output of the XOR logic, and a multiplexer configured to, based on the datatypes being integer types, output the output of the multiplication logic, and based on the datatypes being floating-point types, output an output of the rounder.

[0014] According to another aspect of the disclosure, provided is a computer-readable recording medium having recorded thereon a program for performing an operating method of a multiplier in a computer.

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] The above and other aspects, features, and advantages of certain embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

[0016] The disclosure may be easily understood by combinations of the following detailed description and the accompanying drawings, and the reference numerals respectively refer to structural elements.

[0017] FIG. **1** is a block diagram for explaining a configuration of a hardware accelerator according to an embodiment of the disclosure;

[0018] FIG. **2** is a block diagram for explaining a configuration of a multiply-accumulate (MAC) array according to an embodiment of the disclosure;

[0019] FIG. **3** is a block diagram for explaining a configuration of a first array according to an embodiment of the disclosure;

[0020] FIG. **4** is a block diagram for explaining an operation of a MAC array according to an embodiment of the disclosure;

[0021] FIG. **5** is a conceptual diagram for explaining convolution operations according to an embodiment of the disclosure;

[0022] FIG. **6** is a conceptual diagram illustrating an operation of a MAC array that performs an accumulated convolution operation according to an embodiment of the disclosure;

[0023] FIG. **7** is a conceptual diagram illustrating an operation of a first sub-array that performs a non-accumulated convolution operation according to an embodiment of the disclosure;

[0024] FIG. **8** is a conceptual diagram for explaining a configuration and operation of a second sub-array according to an embodiment of the disclosure;

[0025] FIG. **9** is a block diagram for explaining a configuration of a MAC operator according to an embodiment of the disclosure;

[0026] FIG. **10** is a block diagram for explaining the configuration of a multiplier according to an embodiment of the disclosure;

[0027] FIGS. **11**A to **11**C are conceptual diagrams for explaining operations of a multiplier according to an embodiment of the disclosure;

[0028] FIG. **12** is a block diagram for explaining a configuration of multiplication logic according to an embodiment of the disclosure;

[0029] FIG. **13** is a conceptual diagram illustrating a multiplication operation according to an embodiment of the disclosure;

[0030] FIGS. **14**A and **14**B are block diagrams for explaining operations of a multiplication operation according to an embodiment of the disclosure;

[0031] FIG. **15** is a block diagram for explaining a configuration of addition logic according to an embodiment of the disclosure;

[0032] FIG. **16** is a block diagram for explaining a configuration of sub-addition logic according to an embodiment of the disclosure;

[0033] FIG. **17** is a block diagram for explaining a configuration of XOR logic according to an embodiment of the disclosure;

[0034] FIG. **18** is a block diagram for explaining a configuration of an electronic device according to an embodiment of the disclosure;

[0035] FIG. **19** is a flowchart for explaining an operation of a multiplier according to an embodiment of the disclosure;

[0036] FIG. **20** is a flowchart for explaining an operation of shifting logic according to an embodiment of the disclosure;

[0037] FIG. **21** is a flowchart for explaining an operation of a selective adder according to an embodiment of the disclosure.

[0038] FIG. **22** is a flowchart for explaining an operation of addition logic according to an embodiment of the disclosure;

[0039] FIG. **23** is a flowchart for explaining an operation of XOR logic according to an embodiment of the disclosure; and

[0040] FIG. **24** is a flowchart for explaining operations of a normalizer and a rounder according to an embodiment of the disclosure.

DETAILED DESCRIPTION

[0041] Reference will now be made in detail to embodiments, examples of which are illustrated in the accompanying drawings, wherein like reference numerals refer to like elements throughout. In this regard, the present embodiments may have different forms and should not be construed as being limited to the descriptions set forth herein. Accordingly, the embodiments are merely described below, by referring to the figures, to explain aspects of the present description. As used herein, the term "and/or" includes any and all combinations of one or more of the associated listed items. Expressions such as "at least one of," when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list.

[0042] Although terms used herein are of among general terms which are currently and broadly used by considering functions in the disclosure, these terms may vary according to intentions of those of ordinary skill in the art, precedents, the emergence of new technologies, etc. In addition, there may be terms selected arbitrarily by the applicants in particular cases, and in these cases, the meaning of those terms will be described in detail in the corresponding portions of the detailed description. Therefore, the terms used herein should be defined based on the meaning thereof and descriptions made throughout the specification, rather than based on names simply called.

[0043] The singular terms used herein are intended to include the plural forms as well, unless the context clearly indicates otherwise. All terms used herein, including technical and scientific terms, have the same meaning as generally understood by those of ordinary skill in the art.

[0044] It will be understood that, throughout the specification, when a portion is referred to as "comprising" or "including" a structural element, the portion may further include another structural element in addition to the structural element rather than exclude the other structural element, unless otherwise stated. In addition, the term such as " . . . unit", " . . . portion", " . . . module", or the like used herein refers to a unit for processing at least one function or operation, and this may be implemented by hardware, software, or a combination of hardware and software

[0045] The expression "configured (or set) to" used herein may be used interchangeably with, for example, "suitable for", "having the capacity to", "designed to", "adapted to", "made to", or "capable of" depending on the circumstances. The expression "configured (or set) to" does not essentially mean "specially designed in hardware to". Rather, in some circumstances, the expression "system configured to" may mean that the system may perform an operation together with another device or parts. For example, the phrase "processor configured (or set) to perform A, B, and C" may mean a dedicated processor (for example, an embedded processor) for performing the operations, or a generic-purpose processor (for example, a CPU or an application processor) that may perform the operations by executing one or more software programs stored in a memory.

[0046] In addition, throughout the specification, it should be understood that, when a structural element is referred to as being "coupled to" or "connected to" another structural element, the structural element may be directly coupled to or directly connected to the other structural element or may be coupled to or connected to the other structural element with an intervening structural element therebetween, unless otherwise stated.

[0047] In the disclosure, functions related to "artificial intelligence" are operated by a processor and a memory. The processor may include one or more processors. Here, the one or more processors may include a general-purpose processor, such as a CPU, an application processor (AP), or a digital signal processor (DSP), a dedicated graphics processor, such as a graphics processing unit (GPU) or a vision processing unit (VPU), or a dedicated artificial intelligence processor, such as a neural processing unit (NPU). The one or more processors control input data to be processed according to an artificial intelligence model or predefined operation rules stored in a memory. Alternatively, when the one or more processors include dedicated artificial intelligence processors, the dedicated artificial intelligence processors may be designed with a hardware structure specialized in processing a specific artificial intelligence model.

[0048] The artificial intelligence model or the predefined operation rules are characterized by being made by training. Here, "being made by training" means that a basic artificial intelligence model is trained by a learning algorithm by using a large number of pieces of training data, and thus, the artificial intelligence model or the predefined operation rules set to perform intended features (or purposes) are made. Such training may be performed by a device itself in which artificial intelligence according to the disclosure is performed, or may be performed by a separate server and/or system. The learning algorithm may include, but is not limited to, for example, supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning.

[0049] In an embodiment of the disclosure, an "artificial intelligence model" may include a neural network model. The neural network model may include a plurality of neural network layers. Each of the plurality of neural network layers has a plurality of weight values and performs a neural network operation through an operation between an operation result of a previous layer and the plurality of weight values. The plurality of weight values of each of the plurality of neural network layers may be optimized by a training result of the artificial intelligence model. For example, the plurality of weight values may be updated such that a loss value or a cost value obtained by the artificial intelligence model during a training process is minimized. An artificial neural network model may include a deep neural network (DNN), for example, a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted Boltzmann machine (RBM), a deep belied network (DBN), a bidirectional recurrent deep neural network (BRDNN), or a deep Q-network, but the disclosure is not limited thereto.

[0050] In the disclosure, a tensor may denote an n-dimensional array of data. Each dimension of the tensor may form an axis. For example, a 0-dimensional tensor may represent a scalar value, a 1-dimensional tensor may represent a vector value, and a 2-dimensional tensor may represent a matrix. For example, a 3 or more-dimensional tensor may include a plurality of matrices and may form 3 or more axes.

[0051] In the disclosure, a batch may indicate a unit for updating parameters by grouping all datasets for the training of the artificial intelligence model. For example, all the datasets may be grouped into a plurality of batches. The parameters of the artificial intelligence model may be updated for every one batch. One batch may include a predefined number of mini-batches

[0052] In the disclosure, multiply-accumulate (MAC) operators may be arranged in an n×m array in the form of n rows and m columns. The n×m array may perform n×m matrix operations by using the MAC operators.

[0053] In the disclosure, the expression "the processor may perform an A operation" may indicate that the processor may execute at least one instruction corresponding to the A operation.

[0054] Hereinafter, embodiments of the disclosure will be described in detail with reference to the accompanying drawings such that those of ordinary skill in the art may easily implement the embodiments. However, the disclosure may be implemented in different ways and is not limited to the embodiments described herein.

[0055] FIG. **1** is a block diagram for explaining a configuration of a hardware accelerator **100** according to an embodiment of the disclosure.

[0056] Referring to FIG. **1**, the hardware accelerator **1000** may include a multiply-accumulate (MAC) array **1100**, a scratchpad memory **1200**, a direct memory access (DMA) controller **1300**, a central controller **1400**, and high bandwidth memory (HBM) **1500**. However, all of the illustrated structural elements are not essential structural elements. The hardware accelerator **1000** may be implemented with more structural elements than the illustrated structural elements or may be implemented with fewer structural elements.

[0057] The MAC array **1100** may include a plurality of MAC operators. In the disclosure, a MAC operator may also be referred to as a MAC unit. The MAC operator may perform multiplications between input values. The MAC operator (e.g., a first MAC operator) may receive an output (or may also be referred to as a partial sum output) of an adjacent MAC operator (e.g., a second MAC operator). The MAC operator (e.g., the first MAC operator) may sum the output of the adjacent MAC operator (e.g., the second MAC operator) and a multiplication result. The MAC operator (e.g., the first MAC operator) may transfer a summation result to another adjacent MAC operator (e.g., a third MAC operator).

[0058] The MAC array **1100** may perform operations of at least one deep neural network (DNN) model. The operations of the DNN model may be required for learning or inference

of the DNN model. For example, the operations of the DNN model may be divided into DNN operations and non-DNN operations. DNN operations may include non-cumulative operations such as weight gradient operation, depthwise (DW) convolution, dilated convolution, and up convolution, and cumulative operations such as (general) convolution, pointwise convolution, or fully connected layer operation. Non-DNN operations may include operations in non-DNN layers, such as ReLU, batch normalization, and softmax functions in the DNN model.

[0059] In an embodiment of the disclosure, the MAC array **1100** may perform DNN operations by using a plurality of MAC operators. The MAC array **1100** may group the plurality of MAC operators into at least one group. The at least one group may perform different DNN operations. In an embodiment of the disclosure, the MAC operators may be arranged in the MAC array **1100** in the form of 128×128. The specific configuration and operation of the MAC array **1100** will be described in detail with reference to FIG. **2**.

[0060] In an embodiment of the disclosure, unlike shown in FIG. **1**, the hardware accelerator **1000** may include a vector unit. The vector unit may include a plurality of arithmetic and logic units (ALUs). The vector unit may perform non-DNN operations by using the plurality of ALUs.

[0061] According to an embodiment of the disclosure, the MAC array **1100** may perform an operation on a plurality of DNN models or perform an operation on one DNN model in a multitenant environment, by combining the plurality of MAC operators into the at least one group or dividing the at least one group. According to an embodiment of the disclosure, the MAC array **1100** may support the multitenant environment through a flexible structure, thereby increasing hardware resource utilization with respect to the plurality of MAC operators.

[0062] The scratchpad memory **1200** may store data corresponding to operands required for the operation of the DNN model. For example, the scratchpad memory **1200** may store input tensors and/or weight tensors. The scratchpad memory **1200** may perform a function of prefetching data from the external memory **10** through the DMA controller **1300**.

[0063] MAC array **1100** and/or vector unit may fetch data from scratchpad memory **1200**. The scratchpad memory **1200** may perform a function of prefetching data from an external memory **10** through the DMA controller **1300**.

[0064] The MAC array **1100** and/or the vector unit may store data in the scratchpad memory **1200**. The scratchpad memory **1200** may store partial sum outputs calculated from the MAC array **1100** and/or the vector unit.

[0065] According to an embodiment of the disclosure, a compiler of the scratchpad memory **1200** may easily predict a memory access pattern by a deterministic data flow of the DNN operation. However, the disclosure is not limited thereto. In an embodiment of the disclosure, the scratchpad memory **1200** may be omitted, and the hardware accelerator **1000** may include a static random access memory (SRAM) cache.

[0066] The DMA controller **1300** may control data transfer between internal memories (e.g., the scratchpad memory **1200** and/or the HBM **1500**), data transfer between an internal memory and the external memory **10**, and/or data transfer between an internal memory and the MAC array **1100**. The hardware accelerator **1000** may load data and/or programs from the external memory **10** by using the DMA controller **1300**. The hardware accelerator **1000** may transfer data to the external memory **10** by using the DMA controller **1300**.

[0067] The central controller **1400** may control the overall operations of the hardware accelerator **1000**. The central controller **1400** may transfer control signals to the MAC array **1100**, the scratchpad memory **1200**, and the DMA controller **1300**. The central controller **1400** may store program codes or instructions for controlling the hardware accelerator **1000**. The central controller **1400** may support instructions such as interrupt, halt, synchronization, no-operation (NOP), etc. In an embodiment of the disclosure, the central controller **1400** may be implemented as a RISC-V CPU.

[0068] The central controller **1400** may transfer, to the MAC array **1100**, a control signal to combine the MAC operators of the MAC array **1100** into the at least one group or to divide the at least one group. The central controller **1400** may transfer, to the MAC array **1100**, a control signal to activate and/or deactivate at least one operation logic included in a multiplier of the MAC operator.

[0069] The HBM **1500** may store data required for the operation of the DNN model. The capacity of the HBM **1500** may be greater than the capacity of the scratchpad memory **1200**. The HBM **1500** may receive data from the scratchpad memory **1200** or transfer data to the scratchpad memory **1200** through control of the DMA controller **1300**.

[0070] FIG. **1** shows that the hardware accelerator **1000** includes the scratchpad memory **1200** and the HBM **1500** as internal memories, but the disclosure is not limited thereto. For example, the hardware accelerator **1000** may include at least one of a flash memory type memory, a hard disk type memory, a multimedia card micro type memory, a card type memory (for example, an SD or XD memory or the like), random access memory (RAM), static random access memory (SRAM), read-only memory (ROM), electrically erasable programmable read-only memory (EEPROM), programmable read-only memory (PROM), magnetic memory, a magnetic disk, and an optical disk.

[0071] FIG. **2** is a block diagram for explaining the configuration of the MAC array **1100** according to an embodiment of the disclosure. Redundant descriptions given with reference to FIG. **1** are omitted. For convenience of description, it is described that the MAC array **1100** is a MAC array in which 128×128 MAC operators are arranged, but the disclosure is not limited thereto. For example, the MAC array **1100** may be a MAC operator in which m×n MAC operators are arranged. Here, m and n may be natural numbers.

[0072] Referring to FIG. **1** along with FIG. **2**, the MAC array **1100** may include a plurality of arrays **1110** and global multiplexer logic **1120**. Each of the plurality of arrays **1100** may include the same number of MAC operators. For convenience of description, it is described that each of the plurality of arrays **1100** is an array in which 64×64 MAC operators are arranged, but the disclosure is not limited thereto.

[0073] The MAC array **1100** may include four 64×64 arrays **1110_1**, **1110_2**, **1110_3**, and **1110_4**. However, the disclosure is not limited thereto, and the MAC array **1100** may be configured to include smaller and more arrays, or include larger and fewer arrays.

[0074] The global multiplexer logic **1120** may connect or disconnect at least two of the plurality of arrays **1110**. The global multiplexer logic **1120** may receive a first control signal from the central controller **1400**. The first control signal may respond to whether the plurality of arrays **1110** are connected to each other.

[0075] In an embodiment of the disclosure, the first control signal may correspond to an operation mode of the MAC array **1100**. For example, in a first operation mode, the global multiplexer logic **1120** may disconnect the plurality of arrays **1110** from each other. For example, in a second operation mode, the global multiplexer logic **1120** may connect at least two of the plurality of arrays **1110**.

[0076] The connected arrays **1110** may transfer and receive data therebetween. For example, when the first array **1110_1** and the second array **1110_2** are connected to each other, the first array **1110_1** and the second array **1110_2** may perform a 128×64 matrix operation. For example, when the first array **1110_1** and the third array **1110_3** are connected to each other, the first array **1110_1** and the third array **1110_3** may perform a 64×128 matrix operation. For example, when the first array **1110_1**, the second array **1110_2**, the third array **1110_3**, and the fourth array **1110_4** are all connected to each other, the first array **1110_1**, the second array **1110_2**, the third array **1110_3**, and the fourth array **1110_4** may perform a 128×128 matrix operation.

[0077] For example, among the plurality of arrays **1110**, two arrays (e.g., the second array **1110_2** and the fourth array **1110_4**) may be grouped, and the remaining arrays may not be grouped. The two grouped arrays may perform a 128×64 matrix operation or a 64×128 matrix operation. The two grouped arrays may process an operation of a DNN model with a preset high priority, and the two ungrouped arrays may process an operation of a DNN model with a preset low priority.

[0078] The global multiplexer logic **1120** may include a plurality of multiplexers. Each of the plurality of multiplexers may or may not transfer an output of one array (e.g., the first array **1110_1**) among the plurality of arrays **1110** to an adjacent array (e.g., the third array **1120_3**) in response to a first control signal. For example, each of the plurality of multiplexers may transfer an output of one array (e.g., the first array **1110_1**) among the plurality of arrays **1110** to an adjacent array (e.g., the third array **1120_3**) based on the first control signal corresponding to a first logic value. For example, each of the plurality of multiplexers may not transfer an output of one array (e.g., the first array **1110_1**) among the plurality of arrays **1110** to an adjacent array (e.g., the third array **1120_3**) based on the first control signal corresponding to the first logic value.

[0079] In an embodiment of the disclosure, the plurality of arrays **1110** may be grouped into at least one array group by the global multiplexer logic **1120**. The MAC array **1100** may perform operations on a plurality of tenants. Each of the at least one array group may perform operations on different single tenants among the plurality of tenants.

[0080] The plurality of multiplexers may be connected between a plurality of MAC operators disposed in a last row, a last column, a first row, or a first column of the arrays.

[0081] In an embodiment of the disclosure, the MAC operators arranged in a last row of the first array **1110_1** may be respectively connected to the plurality of multiplexers. The corresponding multiplexers may be respectively connected to the MAC operators arranged in a first row of the third array **1110_3**.

[0082] In an embodiment of the disclosure, the MAC operators arranged in a last column of the first array **1110_1** may be respectively connected to the plurality of multiplexers. The corresponding multiplexers may be respectively connected to the MAC operators arranged in a first column of the second array **1110_2**.

[0083] In an embodiment of the disclosure, the MAC operators arranged in a last column of the third array **1110_3** may be respectively connected to the plurality of multiplexers. The corresponding multiplexers may be respectively connected to the MAC operators arranged in a first column of the fourth array **1110_4**.

[0084] In an embodiment of the disclosure, the MAC operators arranged in a last row of the second array **1110_2** may be respectively connected to plurality of multiplexers. The corresponding multiplexers may be respectively connected to the MAC operators arranged in a first row of the fourth array **1110_4**.

[0085] FIG. **3** is a block diagram for explaining the configuration of the first array **1110_1** according to an embodiment of the disclosure. Redundant descriptions given with reference to FIGS. **1** and **2** are omitted.

[0086] Referring to FIGS. **1** and **2** along with FIG. **3**, the first array **1110_1** may include a plurality of sub-arrays including a plurality of first sub-arrays **1111** and a second sub-array **1112**. The first array **1110_1** may include local multiplexer logic **1113**.

[0087] In an embodiment of the disclosure, MAC operators may be arranged in each of the plurality of first sub-arrays **1111** in the form of 9×64. In an embodiment of the disclosure, the number of first sub-arrays **1111** may be 7. For example, the first array **1110_1** may include seven first sub-arrays **1111_1** to **1111_7**.

[0088] In an embodiment of the disclosure, MAC operators may be arranged in the second sub-array **1112** in the form of 1×64. The specific structure of the second sub-array **1112** will be described in detail in FIG. **8**.

[0089] The local multiplexer logic **1113** may be disposed between the plurality of sub-arrays. For example, local multiplexer logics **1113_1** to **1113_7** may be disposed between the first sub-arrays **1111_1** to **1111_7** and the second sub-array **1112**. The local multiplexer logic **1113** may connect or disconnect the plurality of sub-arrays. For example, the local multiplexer logic **1113_1** may be disposed between the first sub-array **1111_1** and the first sub-array **1111_2**. The local multiplexer logic **1113_1** may connect or disconnect the first sub-array **1111_1** and the first sub-array **1111_2**. Exemplarily, the local multiplexer logic **1113_7** may be disposed between the first sub-array **1111_7** and the second sub-array **1112**. The local multiplexer logic **1113_7** may connect or disconnect the first sub-array **1111_7** and the second sub-array **1112** to or from each other.

[0090] In an embodiment of the disclosure, the local multiplexer logic **1113** may connect or disconnect the plurality of sub-arrays to or from each other based on a second control signal. The local multiplexer logic **1113** may receive the second control signal from the central controller **1400**. The second control signal may correspond to whether the plurality of sub-arrays are connected to each other.

[0091] In an embodiment of the disclosure, the second control signal may correspond to an operation mode of the

first array **1110_1**. For example, in a first operation mode, the local multiplexer logic **1113** may disconnect the plurality of sub-arrays from each other. For example, in a second operation mode, the local multiplexer logic **1113** may connect the plurality of sub-arrays to each other.

[0092] In an embodiment of the disclosure, the plurality of sub-arrays may perform a convolution operation between a first tensor and a second tensor. The plurality of sub-arrays may generate a partial sum tensor as a result of a convolution operation. For example, based on the second control signal indicating the first operation mode of the first array **1110_1**, the first array **1110_1** may accumulate partial sum tensors output by the plurality of sub-arrays. For example, based on the second control signal indicating the second operation mode of the first array **1110_1**, the first array **1110_1** may not accumulate partial sum tensors output by the plurality of sub-arrays.

[0093] In an embodiment of the disclosure, it will be described below under the assumption that the first array **1110_1** operates in the first operation. The first array **1110_1** may perform a 64×64 matrix operation. For example, an output of the first sub-array **1111_1** may be transferred to the first sub-array **1111_2** through the local multiplexer logic **1113_1**. An output of the first sub-array **1111_2** may be transferred to the first sub-array **1111_3** through the local multiplexer logic **1113_2**. An output of the first sub-array **1111_3** may be transferred to the first sub-array **1111_4** through the local multiplexer logic **1113_3**. An output of the first sub-array **1111_4** may be transferred to the first sub-array **1111_5** through the local multiplexer logic **1113_4**. An output of the first sub-array **1111_5** may be transferred to the first sub-array **1111_6** through the local multiplexer logic **1113_5**. An output of the first sub-array **1111_6** may be transferred to the first sub-array **1111_7** through the local multiplexer logic **1113_6**. An output of the first sub-array **1111_7** may be transferred to the second sub-array **1112** through the local multiplexer logic **1113_7**.

[0094] In an embodiment of the disclosure, it will be described below under the assumption that the first array **1110_1** operates in the second operation mode. Each of the outputs of the first sub-arrays **1111_1** to **1111_7** and the second sub-array **1112** may be transferred to the scratchpad memory **1200**.

[0095] For convenience of description, the first array **1110_1** has been described as an example, but the configurations, functions, and operations of the plurality of arrays **1100** (e.g., the second array **1110_2**, the third array **1110_3**, and the fourth array **1110_4** in FIG. **2**) included in the MAC array (**1100** in FIG. **2**) may respectively correspond to the configuration, function, and operation of the first array **1110_1**.

[0096] FIG. **4** is a block diagram for explaining the operation of the MAC array **1100** according to an embodiment of the disclosure. Redundant descriptions given with reference to FIGS. **1** to **3** are omitted. For convenience of description, at least some of components shown in FIGS. **1** to **3** are omitted in FIG. **4**.

[0097] Referring to FIGS. **1** to **3** along with FIG. **4**, the MAC array **1100** may include a plurality of MAC operators. The MAC operator may transfer a partial sum output to another MAC operator (e.g., a lower MAC operator) in a column direction. The MAC operator may forward data of an input tensor to another MAC operator (e.g., a right MAC operator) in a row direction.

[0098] The MAC array **1100** may include the first array **1110_1**, the second array **1110_2**, and the global multiplexer logic **1120**. Multiplexers of the global multiplexer logic **1120** may be disposed between MAC operators in a last row of the first array **1110_1** and MAC operators in a first row of the second array **1110_2**. For example, outputs (e.g., forwarded data) of the MAC operators in the last row of the first array **1110_1** may be respectively transferred to the multiplexers of the global multiplexer logic **1120**.

[0099] Each of the MAC operators in the last row of the first array **1110_1** may correspond to one multiplexer. By an operation of one multiplexer that operates based on a logic value of a first control signal, the output of each of the MAC operators in the last row of the first array **1110_1** or data from the scratchpad memory **1200** may be transferred to each of the MAC operators in the first column of the second array **1110_2**. For example, when the first control signal has the first logical value (e.g., '0'), the outputs (e.g., forwarded data) of the MAC operators in the last row of the first array **1110_1** may be respectively transferred to the MAC operators in the first column of the second array **1110_2**. For example, when the first control signal has a second logic value (e.g., '1'), the data from the scratchpad memory **1200** may be transferred to each of the MAC operators in the first column of the second array **1110_2**.

[0100] The first array **1110_1** may include the first sub-array **1111_1**, the first sub-array **1111_2**, and the local multiplexer logic **1113_1**. MAC operators in a last row of the first sub-array **1111_1** may be respectively connected to multiplexers of the local multiplexer logic **1113_1**. Outputs (e.g., partial sum outputs) of the MAC operators in the last row of the first sub-array **1111_1** may be respectively connected to multiplexers of the local multiplexer logic **1113_1**.

[0101] Each of the MAC operators in a last row of the first sub-array **1111_1** may correspond to two multiplexers. By operations of the two multiplexers operating based on a logic value of the second control signal, the output of each of the MAC operators in the last row may be transferred to the scratchpad memory **1200** or to each of MAC operators in a first row of the first sub-array **1111_2**. For example, when the second control signal has a first logic value (e.g., '0'), the output of the MAC operator may be transferred to the scratchpad memory **1200**. For example, when the second control signal has a second logic value (e.g., '1'), the output of the MAC operator may be transferred to the first sub-array **1111_2**.

[0102] The second array **1110_2** may include a first sub-array **1111_8** and a first sub-array **1111_9**. The configurations, operations, and functions of the first sub-array **1111_8**, the first sub-array **1111_9**, and the local multiplexer logic **1113_8** may respectively correspond to the configurations, operations, and functions of the first sub-array **1111_1**, the first sub-array **1111_2**, and the local multiplexer logic **1113_8**.

[0103] FIG. **5** is a conceptual diagram for explaining convolution operations according to an embodiment of the disclosure.

[0104] Referring to FIG. **1** along with FIG. **5**, the MAC array **1100** may perform convolution operations between an input tensor and a weight tensor. According to types of convolution operations, the MAC array **1100** may obtain a first output that accumulates partial sum tensors that are results of a convolution operation between the input tensor

and the weight tensor, or may obtain a second output that does not accumulate partial sum tensors.

[0105] The convolution operations may be divided into a first operation that accumulates partial sum tensors in a direction of an input channel I.C of the input tensor, and a second operation that does not accumulate partial sum tensors in the direction of the input channel I.C of the input tensor. Examples of operations are shown in Table 1 below.

TABLE 1

Types of operations	Examples
Accumulation? Yes	Conventional convolution, Fully-connected layer, GEMM, GEMV
Accumulation? No	Depthwise convolution, Dilated convolution, Weight gradient ($\partial\mathcal{L}/\partial W$) in the training process of convolutions

[0106] Referring to Table 1, for example, when a weight gradient operation, DW convolution, or dilated convolution, etc. in training of a DNN model is performed, the MAC array **1100** may perform a first operation. For example, when performing a (general) convolution in which operation results are accumulated, fully connected layer operation, general matrix-matrix multiplication (GEMM) operation, or general matrix vector multiplication (GEMV) operation, the MAC array **1100** may perform the first operation.

[0107] In an embodiment of the disclosure, the MAC array **1100** may identify which operation is required at each step of an operation of at least one DNN model being operated. The MAC array **1100** may selectively accumulate and output operation results according to results of identification.

[0108] FIG. **6** is a conceptual diagram illustrating the operation of a MAC array that performs an accumulated convolution operation according to an embodiment of the disclosure. Redundant descriptions given with reference to FIGS. **1** to **5** are omitted.

[0109] Referring to FIG. **6**, input training data may include a plurality of batches. Each of the plurality of batches may include an input tensor, which is a three-dimensional (3D) tensor with a depth of the input channel I.C. Data of input tensors corresponding to the plurality of batches may be prefetched to the MAC array in the form of "dataflow algorithm of oblique input streaming".

[0110] When the accumulated convolution operation is performed in a direction of the input channel I.C of the input tensor, data in the direction of the input channel I.C of the input tensor may be mapped in a column direction of MAC operators of the MAC array.

[0111] Weight values that perform the convolution operation on input training data may include weight tensors, which are 3D tensors with a depth of the input channel I.C. The number of weight tensors may be equal to the number of output channels O.C. When the accumulated convolution operation is performed in the direction of the input channel I.C of the input tensor, data in a direction of the output channel O.C of the weight tensor may be mapped in a row direction of the MAC operators of the MAC array.

[0112] FIG. **6** shows a 4×4 array, but referring to FIG. **2** along with FIG. **6**, data of the input channel I.C. may be input in the column direction of the plurality of arrays **1110** in the form of 64×64, and data of the output channel O.C may be input in the row direction.

[0113] FIG. **7** is a conceptual diagram illustrating the operation of a first sub-array that performs a non-accumulated convolution operation according to an embodiment of the disclosure. Redundant descriptions given with reference to FIGS. **1** to **6** are omitted.

[0114] Referring to FIG. **7**, MAC operators may be disposed in the sub-array (e.g., the first sub-array **1111_1** in FIG. **3**) in the form of 9×64. When the non-accumulated convolution operation is performed in a direction of the input channel I.C of an input tensor, elements of a weight filter may be mapped in a column direction of the sub-array, and data in a direction of the output channel O.C may be mapped in a row direction. The elements of the weight filter may be prefetched in the column direction of the sub-array, and the data in the direction of the output channel O.C may be prefetched in the row direction. For example, the weight filter may correspond to one channel of a weight tensor, which is a 3D tensor.

[0115] Similar to that described with reference to FIG. **6**, training data may include input tensors corresponding to a plurality of batches. The input tensors may be input to the sub-array in the form of "dataflow algorithm of oblique input streaming".

[0116] Because partial sums are not accumulated in the direction of the input channel I.C, a final output from one column of the sub-array may not be transferred to a MAC operator of another sub-array, but transferred to a memory (e.g., the scratchpad memory (**1200** in FIG. **1**)).

[0117] FIG. **8** is a conceptual diagram for explaining the configuration and operation of the second sub-array **1112** according to an embodiment of the disclosure. Redundant descriptions given with reference to FIGS. **1** to **8** are omitted.

[0118] Referring to FIG. **3** along with FIG. **8**, the second sub-array **1112** may include 64 MAC operators MAC**0** to MAC**63**.

[0119] When an accumulated operation is performed in a direction of an input channel, the second sub-array **1112** may not be internally divided and may perform a 64×64 matrix operation together with the first sub-arrays **1111**.

[0120] When a non-accumulated operation is performed in the direction of the input channel, the second sub-array **1112** may be divided into a plurality of groups. For example, the second sub-array **1112** may be divided into seven groups. The second sub-array **1112** may be divided into first to seventh groups **1112***a* to **1112***g*. Each of the first to seventh groups **1112***a* to **1112***g* may perform and output 9 MAC operations.

[0121] For convenience of description, circuit components arranged in the first group **1112***a* of the second sub-array **1112** will be mainly described below. The multiplexer MUX**01** may output one of an output of the first sub-array **1111_7** or data loaded from the scratchpad memory **1200** based on a second control signal. For example, in response to the second control signal having a first logical value (e.g., '0'), the multiplexer MUX**01** may output the data loaded from the scratchpad memory **1200**. For example, in response to the second control signal having a second logic value (e.g., '1'), the multiplexer MUX**01** may output an output of the first sub-array **1111_7**.

[0122] For example, when the second control signal has the first logic value (e.g., '0'), the non-accumulated operation may be performed in the direction of the input channel, and when the second control signal has the second logic

value (e.g., '1')), the accumulated operation may be performed in the direction of the input channel.

[0123] The configuration, function, and operation of the multiplexer MUX**01** may correspond to the configuration, function, and operation of each of the other multiplexers (e.g., MUX**11**, MUX**81**, MUX**631**, etc.) connected to the other MAC operators MAC**1** to MAC**63**.

[0124] The MAC operator MAC**0** may be connected to the multiplexer MUX**01**. The MAC operator MAC**0** may perform a multiplication operation based on an output of the multiplexer MUX**01** and data of an input tensor. The data of the input tensor may be forwarded to other adjacent MAC operators (e.g., MAC**1**). The configuration, function, and operation of the MAC operator MAC**0** may correspond to the configuration, function, and operation of each of the other MAC operators (e.g., MAC**1**, MAC**8**, MAC**63**, etc.).

[0125] A multiplexer MUX**02** may be connected to an adder ADD**0**. The multiplexer MUX**02** may transfer a result of the multiplication operation of the MAC operator MAC**0** to the adder ADD**0** based on the second control signal. The configuration, function, and operation of the multiplexer MUX**02** may correspond to the configuration, function, and operation of each of similarly arranged multiplexers (e.g., MUX**12**, MUX**82**, MUX**632**, etc.).

[0126] The adder ADD**0** may be connected to a register REG**0**. The adder ADD**0** may sum a value stored in the register REG**0** and an output value of the multiplexer MUX**02**. The configuration, function, and operation of the adder ADD**0** may correspond to the configuration, function, and operation of each of similarly arranged adders (e.g., ADD**1**, MUX**8**, MUX**63**, etc.).

[0127] The register REG**0** may be connected to a multiplexer MUX**04**. The register REG**0** may store an output value of the adder ADD**0**. The register REG**0** may transfer the output value of the adder ADD**0** to the multiplexer MUX**04**. The configuration, function, and operation of the register REG**0** may correspond to the configuration, function, and operation of each of similarly arranged registers (e.g., REG**1**, REG**8**, REG**63**, etc.).

[0128] The multiplexer MUX**04** may or may not transfer the stored value of the register REG**0** to the scratchpad memory **1200** based on a fourth control signal. The configuration, function, and operation of the multiplexer MUX**04** may correspond to the configuration, function, and operation of each of similarly arranged multiplexers (e.g., MUX**14**, MUX**84**, MUX**634**, etc.).

[0129] A multiplexer MUX**03** may be connected to a register REG**02**. The multiplexer MUX**03** may transfer the result of the multiplication operation of the MAC operator MAC**0** to the register REG**02** based on the second control signal. The configuration, function, and operation of the multiplexer MUX**03** may correspond to the configuration, function, and operation of each of similarly arranged multiplexers (e.g., MUX**13**, MUX**83**, etc.).

[0130] The register REG**02** may store an output value of the multiplexer MUX**03**. The output value stored in the register REG**02** and an output value of a multiplexer MUX**13** of the adjacent MAC operator MAC**1** may be summed. A summation value may be summed back to an output value of a multiplexer MUX**23** of the next adjacent MAC operator MAC**2**.

[0131] When a non-accumulated operation is performed in the direction of the input channel, by the functions and operations of the above-described circuit elements, outputs of the nine MAC operators may be summed, and a summation value may be transferred to the scratchpad memory **1200**. For example, in the case of the first group **1112***a*, a value stored in a register REG**72** and an output of a multiplexer MUX**83** may be summed and transferred to the scratchpad memory **1200**.

[0132] The configuration, operation, and function of each of the second to seventh groups **1112***b* to **1112***g* correspond to the configuration, operation, and function of the above-described first group **1112***a*, and thus, descriptions thereof are omitted.

[0133] When the MAC operator MAC**63** that does not belong to the first to seventh groups **1112***a* to **1112***g* performs the non-accumulated operation in the direction of the input channel, data may not be mapped. However, when performing the accumulated operation in the direction of the input channel, the MAC operator MAC**63** may transfer a result value to the scratchpad memory **1200** by using a multiplexer MUX**632**, an adder ADD**63**, a register REG**631**, and a multiplexer MUX**634**.

[0134] FIG. **9** is a block diagram for explaining the configuration of a MAC array **900** according to an embodiment of the disclosure. Redundant descriptions given with reference to FIGS. **1** to **8** are omitted. Referring to FIGS. **1** to **8** together, the configuration, operation, and function of each of MAC operators included in the MAC array **1100**, the plurality of arrays **1110**, the first sub-arrays **1111**, and the second sub-array **1112** may respectively correspond to the configuration, operation, and function of the MAC operator **900** of FIG. **9**.

[0135] The MAC operator **900** may include a multiplier **910**, an adder ADD, a precision converter **920**, an adder tree **930**, multiplexers MUX**1** to MUX**5**, and registers REG_I, REG_W, and REG_O.

[0136] The register REG_I may receive an input value. For example, the input value may include bits with a predefined bit width. The register REG_I may store the input value. The register REG_I may be connected to the multiplier **910**. The register REG_I may transfer the stored input value to the multiplier **910**. The register REG_I may be connected to a multiplier of an adjacent MAC operator to forward the stored input value to the adjacent MAC operator.

[0137] The register REG_W may receive a weight value. For example, the weight value may include bits with a predefined bit width. The register REG_W may store the weight value. The register REG_W may be connected to the multiplier **910**. The register REG_W may transfer the stored weight value to the multiplier **910**.

[0138] The multiplier **910** may load the input value from the register REG_I and the weight value from the register REG_W. The multiplier **910** may perform a multiplication operation between the input value and the weight value. The multiplier **910** may be connected to the multiplexer MUX**1** and the multiplexer MUX**2**. The multiplier **910** may support various precisions and datatypes. According to an embodiment of the disclosure, the multiplier **910** may operate flexibly according to precisions and datatypes of the input value and the weight value. The specific structure and operation of the multiplier **910** will be described in detail with reference to FIG. **10**.

[0139] The multiplexer MUX**1** may or may not transfer an output of the multiplier **910** to the adder ADD based on a third control signal. Here, the third control signal may correspond to whether the datatypes of the input value and

the weight value integer types or floating-point types. For example, when the third control signal has a first logical value (e.g., '0'), the datatypes of the input value and the weight value may be floating-point types, and when the third control signal has a second logical value (e.g., '1'), the datatypes of the input value and the weight value may be integer types.

[0140] For example, when the third control signal has the first logic value (e.g., '0'), the multiplexer MUX**1** may not transfer the output of the multiplier **910** to the adder ADD. For example, when the third control signal has the second logic value (e.g., '1'), the multiplexer MUX**1** may transfer the output of the multiplier **910** to the adder ADD.

[0141] The multiplexer MUX**2** may or may not transfer the output of the multiplier **910** to the precision converter **920** based on the third control signal. For example, when the third control signal has the first logic value (e.g., '0'), the multiplexer MUX**1** may transfer the output of the multiplier **910** to the precision converter **920**. For example, when the third control signal has the second logic value (e.g., '1'), the multiplexer MUX**1** may not transfer the output of the multiplier **910** to the precision converter **920**.

[0142] The multiplexer MUX**3** and the multiplexer MUX**4** may receive a first partial sum, which is a partial sum output of the adjacent MAC operator. The multiplexer MUX**3** and MUX**4** may transfer the first partial sum to other structural elements based on the third control signal. For example, when the third control signal has the first logical value (e.g., '0'), the multiplexer MUX**4** may transfer the first partial sum to the adder tree **930**. For example, when the third control signal has the second logic value (e.g., '1'), the multiplexer MUX**3** may transfer the first partial sum to the adder ADD.

[0143] The precision converter **920** may convert precision of the output of the multiplier **910**. For example, when the input value and the weight value are of floating-point types, a result of multiplication performed by the multiplier **910** may be converted to floating-point type 32-bit data through the precision converter **920**. According to an embodiment of the disclosure, accumulated errors of multiplication results may be prevented through the function of the precision converter **920**. The precision converter **920** may convert the output of the multiplier **910** to have a predefined bit width. The precision converter **920** may transfer a converted value to the adder tree **930**. The adder tree **930** may output a second partial sum by summing the first partial sum and an output of the precision converter **920**.

[0144] The adder ADD may sum the output of the multiplier **910** and the first partial sum. The multiplexer MUX**5** may transfer the second partial sum to another adjacent MAC operator by outputting an output of the adder ADD or an output of the adder tree **930**. For example, when the third control signal has the first logical value (e.g., '0'), the multiplexer MUX**5** may output the output of the adder tree **930**. For example, when the third control signal has the second logic value (e.g., '1'), the multiplexer MUX**5** may output the output of the adder ADD.

[0145] The register REG_O may store an output of the multiplexer MUX**5**. The register REG_O may transfer the stored output value, that is, the second partial sum, to another adjacent MAC operator. The other adjacent MAC operator may load the second partial sum from the register REG_O.

[0146] FIG. **10** is a block diagram for explaining a configuration of a multiplier **2000** according to an embodiment of the disclosure. Redundant descriptions given with reference to FIGS. **1** to **9** are omitted. The configuration, operation, and function of the multiplier **910** of FIG. **9** may respectively correspond to the configuration, operation, and function of the multiplier **2000** of FIG. **10**. However, the disclosure is not limited thereto, and the multiplier **2000** may be disposed in a processor (e.g., a central processing unit (CPU), a graphics processing unit (GPU), a neural processing unit (NPU), etc.) by using various operators other than a MAC operator.

[0147] Referring to FIG. **10**, the multiplier **2000** may include multiplication logic **2100**, addition logic **2200**, XOR logic **2300**, a normalizer **2400**, a rounder **2500**, and a multiplexer MUX. However, all of the illustrated structural elements are not essential structural elements. The multiplier **2000** may be implemented with more structural elements than the illustrated structural elements shown, or may be implemented with fewer structural elements.

[0148] The multiplication logic **2100** may include a plurality of sub-multiplication logics. Four sub-multiplication logics are shown in FIG. **10**, but the number of sub-multiplication logics is not limited thereto. The multiplication logic **2100** may output one result value or a plurality of result values based on a first input value and a second input value by using the sub-multiplication logics.

[0149] In an embodiment of the disclosure, the multiplication logic **2100** may be implemented as a carry-save multiplier (CSM), but the disclosure is not limited thereto.

[0150] In an embodiment of the disclosure, the multiplication logic **2100** may perform an 8b×8b multiplication operation, and each of the sub-multiplication logics may perform a 4b×4b multiplication operation, but the disclosure is not limited thereto. In this case, the multiplication logic **2100** may perform one 8b×8b multiplication operation, two 4b×8b operations, two 8b×4b operations, or four 4b×4b operations.

[0151] Different data may be mapped to the multiplication logic **2100** according to precisions and datatypes of the first input value and the second input value. For example, the multiplication logic **2100** may perform multiplication between the first input value and the second input value based on datatypes of the first input value and the second input value being integer types. For example, the multiplication logic **2100** may perform multiplication between a mantissa of the first input value and a mantissa of the second input value, based on datatypes of the first input value and the second input value being floating-point types.

[0152] The multiplication logic **2100** may be connected to the normalizer **2400** and the multiplexer MUX. The multiplication logic **2100** may transfer an operation result to the normalizer **2400** or to the multiplexer MUX.

[0153] The addition logic **2200** may include at least one sub-addition logic. Four sub-addition logics are shown in FIG. **10**, but the number of at least one sub-addition logic is not limited thereto. The at least one sub-addition logic may respectively correspond to at least one output value of the multiplication logic **2100**. In an embodiment of the disclosure, the number of at least one sub-addition logic may correspond to the maximum number of outputs of the multiplication logic **2100**. The at least one sub-addition logic may be arranged in the same number as the maximum number of at least one output of the multiplication logic **2100**. Each of the at least one sub-addition logic may be

activated or deactivated according to the datatypes and precisions of the first input value and the second input value.

[0154] The addition logic **2200** may be deactivated based on the datatypes of the first input value and the second input value being integer types. The addition logic **2200** may sum an exponent of the first input value and an exponent of the second input value based on the datatypes of the first input value and the second input value being floating-point types. Each of the first input value and the second input value may include a plurality of floating-point values. In this case, the addition logic **2200** may perform summation between an exponent of one of the plurality of floating-point values of the first input value and an exponent of one of the plurality of floating-point values of the second input value.

[0155] The addition logic **2200** may subtract a bias value mapped to a bit width of the exponent of the first input value and the exponent of the second input value from a summation result. The bias value may correspond to bit widths of the precisions or exponents of the first and second input values. For example, the bias value may be $2^{(bit\ width\ of\ exponent)}-1$.

[0156] The addition logic **2200** may be connected to the normalizer **2400**. The addition logic **2200** may transfer the operation result to the normalizer **2400**.

[0157] The XOR logic **2300** may perform an exclusive OR operation. The XOR logic **2300** may be deactivated based on the datatypes of the first input value and the second input value being integer types. The XOR logic **2300** may perform an XOR operation between a sign of the first input value and a sign of the second input value, based on the datatypes of the first input value and the second input value being floating-point types.

[0158] The XOR logic **2300** may include at least one sub-XOR logic. Four sub-XOR logics are shown in FIG. **10**, but the number of at least one sub-XOR logic is not limited thereto. The at least one XOR addition logic may respectively correspond to the at least one output value of the multiplication logic **2100**. In an embodiment of the disclosure, the number of at least one sub-XOR logic may correspond to the maximum number of outputs of the multiplication logic **2100**. The at least one sub-XOR logic may be arranged in the same number as the maximum number of at least one output of the multiplication logic **2100**. Each of the at least one sub XOR logic may be activated or deactivated according to the datatypes and precisions of the first input value and the second input value.

[0159] The XOR logic **2300** may be connected to the normalizer **2400**. The XOR logic **2300** may transfer an operation result to the normalizer **2400**.

[0160] The normalizer **2400** may normalize and output input values. The normalizer **2400** may be deactivated based on the datatypes of the first input value and the second input value being integer types. The normalizer **2400** may normalize the output of the multiplication logic **2100** and the output of the addition logic **2200** based on the datatypes of the first and second input values being floating-point types.

[0161] In an embodiment of the disclosure, the normalizer **2400** may include a plurality of sub-normalizers. FIG. **10** shows that the multiplication logic **2100** performs an 8b×8b operation and the normalizer **2400** includes five sub-normalizers, but the disclosure is not limited thereto. For example, one sub-normalizer may correspond to an 8b×8b multiplication operation result, and each of four sub-normalizers may correspond to a 4b×4b multiplication operation result. For example, when the multiplier **2000** performs one 8b×8b floating-point operation, one pre-specified sub-normalizer may be activated, and four unspecified sub-normalizers may be deactivated. For example, when the multiplier **2000** performs four 4b×4b floating-point operations, four pre-specified sub-normalizers may be activated, and one unspecified sub-normalizer may be deactivated.

[0162] The rounder **2500** may round off and output input values with a predefined bit width. The rounder **2500** may be deactivated based on the datatypes of the first input value and the second input value being integer types. The rounder **2500** may round off the input values with the predefined bit width by using the output of the normalizer **2400** and the output of the XOR logic **2300**, based on the datatypes of the first and second input values being floating-point types. FIG. **10** shows that the multiplication logic **2100** performs an 8b×8b operation and the rounder **2500** includes five sub-rounders, but the disclosure is not limited thereto. Whether the sub-rounders are activated corresponds to whether the sub-normalizers of the normalizer **2400** are activated, and thus, a description thereof is omitted.

[0163] The multiplexer MUX may output the output of the multiplication logic **2100** based on the datatypes of the first input value and the second input value being integer types, and output an output of the rounder **2500** based on the datatypes of the first input value and the second input value being integer types.

[0164] In an embodiment of the disclosure, the multiplier **2000** may be controlled by a processor (not shown). The processor may obtain the first input value and the second input value. The processor may identify the datatypes and precisions of the first input value and the second input value. The processor may control each structural element of the multiplier **2000** based on the identified datatypes and precisions.

[0165] In an embodiment of the disclosure, the multiplier **2000** may perform an N-bit×M-bit multiplication operation. In this case, each of the sub-multiplication logics of the multiplication logic **2100** may perform a (N/p)-bit×(M/q)-bit multiplication operation. In this case, N and M may be natural numbers divided by p and q, respectively. In this case, the multiplier **2000** may support multiplication operations such as (N/p)-bit×(M/q)-bit, (2N/p)-bit×(M/q)-bit, . . . , N-bit×(M/q)-bit, N-bit×(2M/q)-bit, . . . , N-bit×M-bit.

[0166] FIGS. **11**A to **11**C are conceptual diagrams for explaining operations of the multiplier **2000** according to an embodiment of the disclosure. Redundant descriptions given with reference to FIG. **10** are omitted. FIGS. **11**A to **11**C show that the multiplier **2000** supports an 8b×8b multiplication operation, but the disclosure is not limited thereto, and may support multiplication operations with respect to various bit widths.

[0167] Referring to FIG. **11**A, it is assumed that a first input value and a second input value have a datatype and precision of bfloat16. bloat16 is a floating-point type, with 1 bit corresponding to a sign, 8 bits corresponding to an exponent, and 7 bits corresponding to a mantissa. Because bloat16 is the floating-point type, the addition logic **2200**, the XOR logic **2300**, the normalizer **2400**, and the rounder **2500** including the multiplication logic **2100** may be activated.

[0168] All sub-multiplication logics of the multiplication logic **2100** may be activated. Because the mantissa additionally includes 1 bit as an implicit bit, an operation may be

performed on the mantissa with 8 bits rather than 7 bits. The multiplication logic **2100** may output a multiplication operation between the 8-bit mantissa of the first input value and the 8-bit mantissa of the second input value.

[0169] Because the multiplication logic **2100** outputs one 8b×8b operation result, one of the sub-addition logics of the addition logic **2200** may be activated. The addition logic **2200** may perform exponential addition between the 8-bit exponent of the first input value and the 8-bit exponent of the second input value, receive a bias value of $2^{(8)}-1$, and output a value obtained by subtracting the bias value from an exponential addition result.

[0170] Because the multiplication logic **2100** outputs one 8b×8b operation result, one of sub-XOR logics of the XOR logic **2300** may be activated. The XOR logic **2300** may perform an XOR operation between the 1-bit sign of the first input value and the 1-bit sign of the second input value.

[0171] Because the multiplication logic **2100** outputs one 8b×8b operation result, one of sub-normalizers of the normalizer **2400** may be activated. The normalizer **2400** may perform normalization on an output of the multiplication logic **2100** and an output of the addition logic **2200**.

[0172] Because the multiplication logic **2100** outputs one 8b×8b operation result, one of sub-rounders of the rounder **2500** may be activated. The rounder **2500** may receive an output of the normalizer **2400** and an output of the XOR logic **2300**. The rounder **2500** may round off the output of the normalizer **2400** with a predefined bit width. The rounder **2500** may connect the output of the XOR logic **2300** to a rounded value and transfer an output to the multiplexer MUX.

[0173] The multiplexer MUX may output an output of the rounder **2500** based on the datatypes of the first input value and the second input value being floating-point types.

[0174] Referring to FIG. **11**B, it is assumed that the first input value and the second input value have a datatype and precision of FP8. FP8 is a floating-point type, with 1 bit corresponding to a sign, 4 bits corresponding to an exponent, and 3 bits corresponding to a mantissa. Accordingly, the first input value may include four FP8 floating-point values, and the second input value may include four FP8 floating-point values. Because FP8 is the floating-point type, the addition logic **2200**, the XOR logic **2300**, the normalizer **2400**, and the rounder **2500** including the multiplication logic **2100** may be activated.

[0175] All sub-multiplication logics of the multiplication logic **2100** may be activated. Because the mantissa additionally includes 1 bit as an implicit bit, an operation may be performed on the mantissa with 4 bits rather than 3 bits. Each of the activated sub-multiplication logics may output a multiplication operation between the 4-bit mantissa of each of the four FP8 values included in the first input value and the 4-bit mantissa of each of the four FP8 values included in the second input value.

[0176] Because the multiplication logic **2100** outputs four 4b×4b operation results, four of sub-addition logics of the addition logic **2200** may be activated. Each of the activated sub-addition logics may perform exponential addition between the 4-bit exponent of each of the four FP8 values included in the first input value and the 4-bit exponent of each of the four FP8 values included in the second input value, receive a bias value of $22^{(4)}-1$, and output a value obtained by subtracting the bias value from an exponential addition result.

[0177] Because the multiplication logic **2100** outputs four 4b×4b operation results, four of sub-XOR logics of the XOR logic **2300** may be activated. Each of the activated sub-XOR logics may perform an XOR operation between the 1-bit sign of each of the four FP8 values included in the first input value and the 1-bit sign of each of the four FP8 values included in the second input value.

[0178] Because the multiplication logic **2100** outputs four 4b×4b operation results, four of the sub-normalizers of the normalizer **2400** may be activated. The normalizer **2400** may perform normalization on an output of the multiplication logic **2100** and an output of the addition logic **2200**.

[0179] Because the multiplication logic **2100** outputs four 4b×4b operation results, four of the sub-rounders of the rounder **2500** may be activated. The rounder **2500** may receive an output of the normalizer **2400** and an output of the XOR logic **2300**. The rounder **2500** may round off the output of the normalizer **2400** with a predefined bit width. The rounder **2500** may connect the output of the XOR logic **2300** to a rounded value and transfer an output to the multiplexer MUX.

[0180] The multiplexer MUX may output an output of the rounder **2500** based on the datatypes of the first input value and the second input value being floating-point types.

[0181] Referring to FIG. **11**C, it is assumed that the first input value and the second input value have a datatype and precision of INT8. INT8 is an integer datatype with a bit width of 8 bits. Because INT8 is an integer type, only the multiplication logic **2100** may be activated, and the addition logic **2200**, the XOR logic **2300**, the normalizer **2400**, and the rounder **2500** may be deactivated.

[0182] All sub-multiplication logics of the multiplication logic **2100** may be activated. The multiplication logic **2100** may output a multiplication operation between the 8-bit integer corresponding to the first input value and the 8-bit integer corresponding to the second input value by using the sub-multiplication logics.

[0183] The multiplexer MUX may output an output of the multiplication logic **2100** based on the datatypes of the first input value and the second input value being integer types.

[0184] FIG. **12** is a block diagram for explaining the configuration of the multiplication logic **2100** according to an embodiment of the disclosure. Redundant descriptions given with reference to FIGS. **10** to **11**C are omitted.

[0185] Referring to FIG. **10** along with FIG. **12**, the multiplication logic **2100** may include four sub-multiplication logics **2110_1**, **2110_2**, **2110_3**, and **2110_4**. However, the disclosure is not limited thereto, and the multiplication logic **2100** may include fewer or more than four sub-multiplication logics.

[0186] Each of the sub-multiplication logics **2110_1**, **2110_2**, **2110_3**, and **2110_4** may be implemented as a 4b×4b multiplier or a 5b×5b multiplier including a sign bit. FIG. **12** shows that each of the sub-multiplication logics **2110_1**, **2110_2**, **2110_3**, and **2110_4** performs the 5b×5b multiplication operation, but the disclosure is not limited thereto.

[0187] Each of the sub-multiplication logics **2110_1**, **2110_2**, **2110_3**, and **2110_4** may include at least one shifter logic SHIFT. The shifter logic may shift an output of each of the sub-multiplication logics **2110_1**, **2110_2**, **2110_3**, and **2110_4** by a predefined number of bits. For example, the shifter logic SHIFT may perform 4-bit left shifting. How-

ever, the disclosure is not limited thereto, and shifting bits and/or shifting direction of the shifter logic SHIFT may vary.

[0188] In an embodiment of the disclosure, two shifter logics may be connected to the first sub-multiplication logic **2110_1**. The two shifter logics may perform left shifting on an output of the first sub-multiplication logic **2110_1** by 8 bits. One shifter logic may be connected to the second sub-multiplication logic **2110_2**. One shifter logic may perform left shifting on an output of the second sub-multiplication logic **2110_2** by 4 bits. One shifter logic may be connected to the third sub-multiplication logic **2110_3**. The one shifter logic may perform left shifting on an output of the third sub-multiplication logic **2110_3** by 4 bits.

[0189] In an embodiment of the disclosure, the multiplication logic **2100** may include a first multiplexer logic **2121** and a second multiplexer logic **2122**. Each of the sub-multiplication logics **2110_1**, **2110_2**, **2110_3**, and **2110_4** may be connected to the first multiplexer logic **2121** and the second multiplexer logic **2122**.

[0190] The first multiplexer logic **2121** may receive a control signal indicating whether summation of the outputs of the sub-multiplication logics **2110_1**, **2110_2**, **2110_3**, and **2110_4** is necessary based on precisions and datatypes of the values input to the multiplication logic **2100**. The first multiplexer logic **2121** may transfer an output of the multiplication logic **2100** to the selective adder ADD based on the control signal indicating that summation of the outputs of the sub-multiplication logics **2110_1**, **2110_2**, **2110_3**, and **2110_4** is necessary. The selective adder ADD may output a first output of 18 bits by summing the outputs of the sub-multiplication logics **2110_1**, **2110_2**, **2110_3**, and **2110_4**.

[0191] The multiplexer logic **2121** may output a second output of 40 bits that is a set of the outputs of the sub-multiplication logics **2110_1**, **2110_2**, **2110_3**, and **2110_4**, based on the control signal indicating that summation of the outputs of the sub-multiplication logics **2110_1**, **2110_2**, **2110_3**, and **2110_4** is unnecessary.

[0192] FIG. **13** is a conceptual diagram illustrating a multiplication operation according to an embodiment of the disclosure.

[0193] Referring to FIG. **12** along with FIG. **13**, it is assumed that a weight value W is 8 bits and an input value X is 8 bits. The weight value W may be divided into a 4-bit first weight sub-word w**0** and a 4-bit second weight sub-word w**1**. The input value X may be divided into a 4-bit first input sub-word x**0** and a 4-bit second input sub-word x**1**. To perform a multiplication operation between 8 bits and 8 bits, a 16-bit result value may be obtained by performing a multiplication operation between 4-bit sub-words and then performing shifting and summation.

[0194] For example, the fourth sub-multiplication logic **2110_4** may obtain a first result value by performing multiplication between the first weight sub-word w**0** and the first input sub-word x**0**. The third sub-multiplication logic **2110_3** may obtain a second result value by performing multiplication between the second weight sub-word w**1** and the first input sub-word x**0**. The second sub-multiplication logic **2110_2** may obtain a third result value by performing multiplication between the first weight sub-word w**0** and the second input sub-word x**1**. The first sub-multiplication logic **2110_1** may obtain a fourth result value by performing multiplication between the second weight sub-word w**1** and the second input sub-word x**1**. A shifter logic may perform left shifting on the second result value and the third result value by 4 bits. The shifter logic may perform left shifting on the fourth result value by 8 bits.

[0195] The multiplication logic **2100** may output a 16-bit result value based on the first result value and shifted second to fourth result values.

[0196] FIGS. **14**A and **14**B are block diagrams for explaining operations of a multiplication operation according to an embodiment of the disclosure. Redundant descriptions given with reference to FIGS. **10** to **13** are omitted.

[0197] Referring to FIG. **14**A, it is assumed that the multiplication logic **2100** performs a multiplication operation between a first input value of INT8 and a second input value of INT8. Each of sub-multiplication logics may perform a multiplication operation between 4-bit sub-words. At least one shifting logic may shift at least a part of the multiplication operation by 4 bits or 8 bits. When a datatype of an input value is INT8, a selective adder is activated, and result values of the sub-multiplication logics may be summed and output.

[0198] In an embodiment of the disclosure, it is assumed that the multiplication logic **2100** performs the multiplication operation between a first input value of bfloat16 and a second input value of bfloat16. Each of the sub-multiplication logics may perform the multiplication operation between 4-bit sub-words among mantissa values of the 8-bit bfloat16. The at least one shifting logic may shift at least a part of the multiplication operation by 4 bits or 8 bits. When the datatype of the input value is bfloat16, the selective adder may be activated, and the result values of the sub-multiplication logic may be summed and output. However, the disclosure is not limited thereto, and the selective adder may be deactivated by predefined settings, regardless of the datatype of the input value.

[0199] Referring to FIG. **14**B, it is assumed that the multiplication logic **2100** performs a multiplication operation between a first input value of INT4 and a second input value of INT4. Each of sub-multiplication logics may perform a multiplication operation between one of a plurality of 4-bit INT4 integer values included in the first input value and one of a plurality of 4-bit INT4 integer values included in the second input value. At least one shifting logic may be deactivated. When the datatype of the input value is IN4, the selective adder is deactivated, and result values of the sub-multiplication logics may be output without being summed.

[0200] In an embodiment of the disclosure, it is assumed that the multiplication logic **2100** performs a multiplication operation between a first input value of FP8 and a second input value of FP8. Each of the sub-multiplication logics may perform a multiplication operation between one of a plurality of 4-bit FP8 mantissa values included in the first input value and one of a plurality of 4-bit FP8 mantissa values included in the second input value. At least one shifting logic may be deactivated. When the datatype of the input value is FP8, the selective adder is deactivated, and result values of the sub-multiplication logics may be output without being summed. However, the disclosure is not limited thereto, and the selective adder may be activated by predefined settings, regardless of the datatype of the input value.

[0201] FIG. **15** is a block diagram for explaining the configuration of the addition logic **2200** according to an

embodiment of the disclosure. Redundant descriptions given with reference to FIGS. **10** to **14**B are omitted.

[0202] Referring to FIG. **10** along with FIG. **15**, the addition logic **2200** may include four sub-addition logics **2210_1**, **2210_2**, **2210_3**, and **2210_4**. However, the disclosure is not limited thereto, and the addition logic **2200** may include fewer or more than four sub-addition logics.

[0203] Each of a first input value and a second input value may include at least one exponent according to a datatype and precision thereof. For example, when datatypes and precisions of the first input value and the second input value are bfloat 16, each of the first input value and the second input value may include one exponent. In this case, only the first sub-addition logic **2210_1** may be activated.

[0204] For example, when the datatypes and precisions of the first input value and the second input value are FP8, each of the first input value and the second input value may include four exponents (e.g., first to fourth exponents). In this case, the first to fourth sub-addition logics **2210_1**, **2210_2**, **2210_3**, and **2210_4** may be activated.

[0205] For convenience of description, the configuration, operation, and function of the first sub-addition logic **2210_1** are mainly described below.

[0206] The first sub-addition logic **2210_1** may include a first unsigned adder ADD**11** and a second unsigned adder ADD**12**. In an embodiment of the disclosure, the first unsigned adder ADD**11** and the second unsigned adder ADD**12** may each be implemented as a CSA, but the disclosure is not limited thereto.

[0207] The first unsigned adder ADD**11** may receive a first exponent of the first input value and a first exponent of the second input value. The first unsigned adder ADD**11** may sum the first exponent of the first input value and the first exponent of the second input value. The first unsigned adder ADD**11** may transfer a summation result to the second unsigned adder ADD**12**.

[0208] The second unsigned adder ADD**12** may subtract a predefined bias value from the summation result. The bias value may vary depending on the precisions and datatypes of the first input value and the second input value. The second unsigned adder ADD**12** may output a subtraction result (e.g., first exponent output).

[0209] The configuration, operation, and function of the first sub-addition logic **2210_1** may respectively correspond to the configuration, operation, and function of each of the second to fourth sub-addition logics **2210_2** to **2210_4**. The configuration, operation, and function of each of the first unsigned adder ADD**11** and the second unsigned adder ADD**12** may respectively correspond to the configuration, operation, and function of each of first unsigned adders ADD**21**, ADD**31**, and ADD**41** and second unsigned adders ADD**22**, ADD**32**, and ADD**42** of the second to fourth sub-addition logics **2210_2** to **2210_4**.

[0210] FIG. **16** is a block diagram for explaining the configuration of the first sub-addition logic **2210_1** according to an embodiment of the disclosure. Redundant descriptions given with reference to FIGS. **10** to **15** are omitted. For convenience of description, the configuration of the first sub-addition logic **2210_1** in FIG. **15** among sub-addition logics is described as an example.

[0211] Referring to FIG. **15** together with FIG. **16**, the first unsigned adder ADD**11** may include a plurality of full adders FA and a half adder HA. FIG. **16** shows that the first unsigned adder ADD**11** includes 7 full adders FA, but the disclosure is not limited thereto, and the first unsigned adder ADD**11** may include fewer or more the 7 full adders FA.

[0212] According to an embodiment of the disclosure, as the number of full adders FA increases, the precision of an exponent that may be processed by the first sub-addition logic **2210_1** may increase. When the sum of the number of full adders FA and the number of half adders HA included in the first unsigned adder ADD**11** is N, the first sub-multiplication logic **2210_1** in FIG. **15** may support an addition operation on exponents of N bits or less. For example, a combination of the plurality of full adders FA and half adders HA shown in FIG. **16** may support addition operations on exponents of 8 bits or less.

[0213] The half adder HA may output a first carry bit C**0** and a first summation bit S**0** by using a first bit EA**0** of a first exponent of a first input value and a first bit EB**0** of a first exponent of a second input value as inputs.

[0214] Each of the plurality of full adders FA may output a next carry bit and a next summation bit by using a previous carry bit, a predefined bit of the first exponent of the first input value, and a predefined bit of the first exponent of the second input value.

[0215] The plurality of full adders FA may include first to seventh full adders. For example, the first full adder may output a second carry bit C**1** and a second summation bit S**1** by using the first carry bit C**0**, a second bit EA**1** of the first exponent of the first input value, and a second bit EB**1** of the first exponent of the second input value as inputs. For example, the second full adder may output a third carry bit C**2** and a third summation bit S**2** by using the second carry bit C**1**, a third bit EA**2** of the first exponent of the first input value, and a third bit EB**2** of the first exponent of the second input value as inputs. For example, the third full adder may output a fourth carry bit C**3** and a fourth summation bit S**3** by using the third carry bit C**2**, a fourth bit EA**3** of the first exponent of the first input value, and a fourth bit EB**3** of the first exponent of the second input value as inputs. For example, the fourth full adder may output a fifth carry bit C**4** and a fifth summation bit S**4** by using the fourth carry bit C**3**, a fifth bit EA**4** of the first exponent of the first input value, and a fifth bit EB**4** of the first exponent of the second input value as inputs. For example, the fifth full adder may output a sixth carry bit C**5** and a sixth summation bit S**5** by using the fifth carry bit C**4**, a sixth bit EA**5** of the first exponent of the first input value, and a sixth bit EB**5** of the first exponent of the second input value as inputs. For example, the sixth full adder may output a seventh carry bit C**6** and a seventh summation bit S**6** by using the sixth carry bit C**5**, a seventh bit EA**6** of the first exponent of the first input value, and a seventh bit EB**6** of the first exponent of the second input value as inputs. For example, the seventh full adder may output an eighth carry bit C**7** and an eighth summation bit S**7** by using the seventh carry bit C**6**, an eighth bit EA**7** of the first exponent of the first input value, and an eighth bit EB**7** of the first exponent of the second input value as inputs.

[0216] The second unsigned adder ADD**12** may include the plurality of full adders FA and the half adder HA. FIG. **16** shows that the second unsigned adder ADD**12** includes 8 full adders FA, but the disclosure is not limited thereto, and the first unsigned adder ADD**11** may include fewer or more the 8 full adders FA. The second unsigned adder ADD**12** may include one more full adder FA than the first unsigned adder ADD**11**.

[0217] The half adder HA of the second unsigned adder ADD**12** may output the first carry bit C**0** and the first summation bit S**0** by using the first summation bit S**0** output by the half adder HA of the first unsigned adder ADD**11** and the first bit B**0** of a bias value as inputs. The first summation bit S**0** output by the half adder HA of the first unsigned adder ADD**11** and the first summation bit S**0** output by the half adder HA of the second unsigned adder ADD**12** may refer to different bits.

[0218] Each of the plurality of full adders FA may output a next carry bit and a next summation bit by using a previous carry bit, a predefined bit of the bias value, and a corresponding summation bit among summation bits output by the full adders FA of the first unsigned adder ADD**11** as inputs.

[0219] The plurality of full adders FA may include eighth to fifteenth full adders. For example, the eighth full adder may output the second carry bit C**1** and the second summation bit S**1** by using the first carry bit C**0**, the second summation bit S**1** output by the first full adder, and a second bit B**1** of the bias value as inputs. For example, the ninth full adder may output the third carry bit C**2** and the third summation bit S**2** by using the second carry bit C**1**, third summation bit S**2** output by the second full adder, and a third bit B**2** of the bias value as inputs. For example, the tenth full adder may output the fourth carry bit C**3** and the fourth summation bit S**3** by using third carry bit C**2**, the fourth summation bit S**3** output by the third full adder, and a fourth bit B**3** of the bias value as inputs. For example, the eleventh full adder may output the fifth carry bit C**4** and the fifth summation bit S**4** by using the fourth carry bit C**3**, the fifth summation bit S**4** output by the fourth full adder, and a fifth bit B**4** of the bias value as inputs. For example, the twelfth full adder may output the sixth carry bit C**5** and the sixth summation bit S**5** by using the fifth carry bit C**4**, the sixth summation bit S**5**, and a sixth bit B**5** of the bias value as inputs. For example, the thirteenth full adder may output the seventh carry bit C**6** and the seventh summation bit S**6** by using the sixth carry bit C**5**, the seventh summation bit S**6**, and a seventh bit B**6** of the bias value as inputs. For example, the fourteenth full adder may output the eighth carry bit C**7** and the eighth summation bit S**7** by using the seventh carry bit C**6**, the eighth summation bit S**7**, and an eighth bit B**7** of the bias value as inputs. For example, the fifteenth full adder may output a ninth summation bit S**8** and a tenth summation bit S**9** by using the eighth carry bit C**7**, the eighth carry bit C**7** output by the seventh full adder, and a ninth bit B**8** of the bias value as inputs.

[0220] The summation bits S**0** to S**9**, which are outputs of the second unsigned adder ADD**12** may be used as output values of the addition logic **2200** with the first exponent of the first input value and the second exponent of the second input value as inputs.

[0221] FIG. **17** is a block diagram for explaining the configuration of the XOR logic **2300** according to an embodiment of the disclosure. Redundant descriptions given with reference to FIGS. **10** to **16** are omitted.

[0222] Referring to FIG. **10** along with FIG. **16**, the XOR logic **2300** may include four sub-XOR logics **2310_1**, **2310_2**, **2310_3**, and **2310_4**. However, the disclosure is not limited thereto, and the XOR logic **2300** may include fewer or more than four sub-XOR logics.

[0223] Each of a first input value and a second input value may include at least one sign according to a datatype and precision thereof. For example, when datatypes and precisions of the first input value and the second input value are bfloat 16, each of the first input value and the second input value may include one sign. In this case, only the first sub-XOR logic **2310_1** may be activated.

[0224] For example, when the datatypes and precisions of the first input value and the second input value are FP8, each of the first input value and the second input value may include four signs (e.g., first to fourth signs). In this case, the first to fourth sub XOR logics **2310_1**, **2310_2**, **2310_3**, and **2310_4** may be activated.

[0225] For convenience of description, the configuration, operation, and function of the first sub-XOR logic **2310_1** are mainly described below.

[0226] The first sub XOR logic **2310_1** may receive a first sign of the first input value and a first sign of the second input value. The first sub XOR logic **2310_1** may perform an XOR operation by using the first sign of the first input value and the first sign of the second input value. For example, when a bit corresponding to the first sign of the first input value and a bit corresponding to the first sign of the second input value are the same, the first sub XOR logic **2310_1** may output the first value, and when the bit corresponding to the first sign of the input value and the bit corresponding to the first sign of the second input value are different, the first sub XOR logic **2310_1** may output the second value. For example, when the first sign of the first input value represents a positive number, and the first sign of the second input value represents a positive number, the first sub-XOR logic **2310_1** may output a sign value representing a positive number. For example, when the first sign of the first input value represents a positive number and the first sign of the second input value represents a negative number, the first sub-XOR logic **2310_1** may output a sign value representing a negative number.

[0227] FIG. **18** is a block diagram for explaining the configuration of an electronic device **3000** according to an embodiment of the disclosure. Redundant descriptions given with reference to FIGS. **10** to **17** are omitted.

[0228] Referring to FIG. **18**, the electronic device **3000** may include a hardware accelerator **3100**, a processor **3200**, a memory **3300**, and an input/output interface **3400**. However, structural elements of the electronic device **3000** are not limited to the above-described examples, and the electronic device **3000** may include more structural elements or fewer structural elements than the above-described structural elements.

[0229] In an embodiment of the disclosure, at least some of the hardware accelerator **3100**, the processor **3200**, the memory **3300**, and the input/output interface **3400** may be implemented in the form of a single chip, and the processor **3200** may include one or more processors. The electronic device **3000** may be a device such as a PC, laptop, or server, as well as a mobile device such as a smartphone.

[0230] The configuration, function, and operation of the hardware accelerator **3100** may respectively correspond to the configuration, function, and operation of the hardware accelerator **1000** described with reference to FIG. **1**. Therefore, redundant descriptions given with reference to FIG. **1** are omitted

[0231] The hardware accelerator **3100** may perform a matrix operation between a first tensor (e.g., an input tensor) and a second tensor (e.g., a weight tensor) by using a plurality of MAC operators.

[0232] The hardware accelerator **3100** may perform learning or inference of a DNN model under the control of the processor **3200**. The hardware accelerator **3100** may read data (e.g., the first tensor and the second tensor) stored in the memory **3300** to perform an operation. An operation result of the hardware accelerator **3100** may be stored in the memory **3300**.

[0233] In an embodiment of the disclosure, the hardware accelerator **3100** may operate in a first operation mode in which operation results of the plurality of MAC operators are output without being accumulated in a direction of an input channel, or the operation results of the plurality of MAC operators are accumulated in the direction of the input channel output and the accumulated operation results are output.

[0234] The processor **3200** is configured to control a series of processes to operate the hardware accelerator **3100** according to the embodiments described with reference to FIGS. **1** to **17**, and may include one or more processors. At this time, the one or more processors may be general-purpose processors such as a CPU, an AP, or a digital signal processor (DSP).

[0235] The processor **3200** may write data to the memory **3300** or read data stored in the memory **3300**, and in particular, execute a program stored in the memory **3300** to process data according to predefined operation rules or a DNN model. In an embodiment, the processor **2320** may control the hardware accelerator **3100** based on DNN information including at least one of the number of layers of the DNN, types of layers, shapes of tensors, dimensions of tensors, operation modes, bit precisions, types of batch normalizations, types of pooling layers, or types of ReLU functions.

[0236] In an embodiment of the disclosure, the processor **3200** may control the hardware accelerator **3100** to perform an operation of at least one DNN based on a user input. The processor **3200** may control the hardware accelerator **3100** to perform an operation of at least one DNN model based on a plurality of user inputs. When an operation on the same DNN model is requested from another user, the processor **3200** may allow the hardware accelerator **3100** to simultaneously perform operations on the same DNN models based on user inputs corresponding to the user request. According to an embodiment of the disclosure, in a multitenant environment, the hardware accelerator **3100** may simultaneously process requests from other users with respect to operations of the same DNN model by grouping or not grouping MAC operators of the MAC array.

[0237] In an embodiment of the disclosure, the hardware accelerator **3100** may include a multiplier **3110**. The configuration, function, and operation of the multiplier **3110** may respectively correspond to the configuration, function, and operation of the multiplier **2000** described with reference to FIG. **10**. Therefore, redundant descriptions given with reference to FIG. **10** are omitted

[0238] An operation of the multiplier **3110** may be controlled by the processor **3200**. The processor **3200** may obtain a first input value and a second input value from the input/output interface **3400** or the memory **3300**. The processor **3200** may identify datatypes and precisions of the first input value and the second input value. For example, the datatype may be an integer type or a floating-point type, but the disclosure is not limited thereto. For example, the precision may indicate a bit width of the first input value or the second input value. For example, the first input value and/or the second input value may be INT4, INT8, bfloat16, FP8, FP32, etc., but the disclosure is not limited thereto, and the first input value and/or the second input value may have various datatypes and/or precisions.

[0239] In an embodiment of the disclosure, the processor **3200** may distribute bits of the first input value and bits of the second input value to sub-multiplication logics of multiplication logic of the multiplier **3100**, based on the identified datatypes and precisions. The processor **3200** may obtain at least one output of the multiplication logic based on outputs of the sub-multiplication logic.

[0240] In an embodiment of the disclosure, the processor **3200** may determine whether to activate at least one shifting logic of the multiplication logic based on the identified datatypes and precisions. The at least one activated shifting logic may shift at least one of the outputs of the sub-multiplication logics.

[0241] In an embodiment of the disclosure, the processor **3200** may determine whether to activate a selective adder of the multiplication logic based on the identified datatypes and precisions. The activated selective adder may sum the outputs of the sub-multiplication logics.

[0242] In an embodiment of the disclosure, the processor **3200** may distribute bits corresponding to a mantissa of the first input value to the sub-multiplication logics, based on the identified datatypes being floating-point types. The processor **3200** may distribute bits corresponding to a mantissa of the second input value to the sub-multiplication logics, based on the identified datatypes being integer types.

[0243] In an embodiment of the disclosure, the processor **3200** may determine whether to activate an addition logic of the multiplier **3110** based on the identified datatypes and precisions. The activated addition logic may sum bits corresponding to an exponent of the second input value and bits corresponding to an exponent of the first input value. The activated addition logic may subtract bias values mapped to the identified datatypes and precisions from a summation result.

[0244] In an embodiment of the disclosure, the processor **3200** may determine whether to activate an XOR logic of the multiplier **3110** based on the identified datatypes and precisions. The activated XOR logic may perform an XOR operation between a bit corresponding to a sign of the first input value and a bit corresponding to a sign of the second input value.

[0245] In an embodiment of the disclosure, the processor **3200** may determine whether to activate a normalizer based on the identified datatypes and precisions. The activated normalizer may normalize at least one output of the multiplication logic and at least one output of the addition logic.

[0246] In an embodiment of the disclosure, the processor **3200** may determine whether to activate a rounder based on the identified datatypes and precisions. The activated rounder may perform rounding with a predefined bit width by using at least one output of the XOR logic and at least one output of the normalizer.

[0247] In an embodiment of the disclosure, the processor **3200** may determine at least one output of the multiplication logic to be the output of the multiplier **3110**, based on the identified datatypes being integer types. The processor **3200** may determine the output of the normalizer to be the output of the multiplier **3110**, based on the fact that the identified datatypes being floating-point types.

[0248] The hardware accelerator **3100**, the multiplier **3110**, and the processor **3200** may perform operations described in the above-described embodiments, and the operations that are described as being performed by the electronic device **3000** in the above-described embodiments may be regarded as being performed by the hardware accelerator **3100**, the multiplier **3110**, and the processor **3200** unless otherwise stated.

[0249] The memory **3300** is configured to store various programs or data, and may include storage media such as ROM, RAM, hard disk, CD-ROM, and DVD, or a combination of storage media. The memory **3300** may not be present separately but may be included in the processor **3200** or the hardware accelerator **3100**. The memory **3300** may be configured as volatile memory, non-volatile memory, or a combination of volatile memory and non-volatile memory. Programs for performing operations performed by the hardware accelerator **3100** or the processor **3200** may be stored in the memory **3300**. The memory **3300** may provide stored data to the hardware accelerator **3100** or the processor **3200** according to a request from the hardware accelerator **3100** or the processor **3200**. In an embodiment, the memory **3300** may store at least one instruction executed by the processor **3200**. The memory **3300** may store parameters or hyperparameters used to learn or infer a DNN.

[0250] The input/output interface **3400** may include an input interface (e.g., a touch screen, a hard button, a microphone, etc.) for receiving control commands or information from a user, and an output interface (e.g., a display panel, a speaker, etc.) for displaying an execution result of an operation under the user's control or a state of the electronic device **3000**. In an embodiment of the disclosure, the electronic device **3000** may receive hyperparameters of the DNN model and/or a user input requesting an operation of the DNN model from the hardware accelerator **3100** through the input/output interface **3400**

[0251] FIG. **19** is a flowchart for explaining the operation of a multiplier according to an embodiment of the disclosure. Redundant descriptions given with reference to FIGS. **10** to **18** are omitted.

[0252] In an embodiment of the disclosure, operations S**1910** to S**1940** may be performed by at least one of the electronic device **3000**, the multiplier **3110**, the hardware accelerator **3100**, and the processor **3200** of FIG. **18**. However, the disclosure is not limited thereto, and operations S**1910** to S**1940** may be performed by any electronic device. An operating method of the multiplier or a controlling method of the multiplier according to an embodiment of the disclosure is not limited to that shown in FIG. **19**, may omit at least one of the operations shown in FIG. **19**, and may further include an operation not shown in FIG. **19**.

[0253] In operation S**1910**, the electronic device **3000** may obtain a first input value and a second input value. In an embodiment of the disclosure, each of the first input value and the second input value may include a plurality of bits having the same datatype and precision. For example, the first input value may correspond to training data or inference target data. The first input value may represent a specific value of an input tensor. For example, the second input value may correspond to weight data. The second input value may represent a specific value of a weight tensor.

[0254] In operation S**1920**, the electronic device **3000** may identify datatypes and precisions of the first input value and the second input value. The electronic device **3000** may transfer a control signal mapped according to the datatypes and precisions of the first input value and the second input value to the multiplier. The multiplier may distribute data to components of the multiplier and/or activate/deactivate the components of the multiplier based on the control signal.

[0255] In operation S**1930**, the electronic device **3000** may distribute bits of the first input value and bits of the second input value to sub-multiplication logics of multiplication logic included in the multiplier, based on the identified datatypes and precisions. In an embodiment of the disclosure, the sub-multiplication logics may perform a 4b×4b multiplication operation that does not process signs, or a 5b×5b multiplication operation that processes signs.

[0256] In operation S**1940**, the electronic device **3000** may obtain at least one output of the multiplication logic based on outputs of the sub-multiplication logics. In an embodiment of the disclosure, the electronic device **3000** may perform shifting on at least one of the outputs of sub-multiplication logics. In an embodiment of the disclosure, the electronic device **3000** may sum the outputs of sub-multiplication logics.

[0257] FIG. **20** is a flowchart for explaining the operation of shifting logic according to an embodiment of the disclosure. Redundant descriptions given with reference to FIGS. **10** to **19** are omitted.

[0258] In an embodiment of the disclosure, operations S**2010** and S**2020** may be performed by at least one of the electronic device **3000**, the multiplier **3110**, the hardware accelerator **3100**, and the processor **3200** of FIG. **18**. However, the disclosure is not limited thereto, and operations S**2010** and S**2020** may be performed by any electronic device. An operating method of the shifting logic or a controlling method of the shifting logic according to an embodiment of the disclosure is not limited to that shown in FIG. **20**, may omit at least one of the operations shown in FIG. **20**, and may further include an operation not shown in FIG. **20**.

[0259] In an embodiment of the disclosure, operations S**2010** and S**2020** may be performed by the electronic device **3000** between operations S**1930** and S**1940** of FIG. **19**. However, the time at which operations S**2010** and S**2020** are performed is not limited thereto.

[0260] In operation S**2010**, the electronic device **3000** may determine whether to activate at least one shifting logic of a multiplication logic based on identified datatypes and precisions. The electronic device **3000** may determine whether to activate the at least one shifting logic mapped to the identified datatypes and precisions. The electronic device **3000** may activate or deactivate the at least one shifting logic that shifts outputs of sub-multiplication logics according to an identification result.

[0261] In operation S**2020**, the electronic device **3000** may determine to activate the at least one shifting logic and activate the at least one shifting logic that shifts at least one of the outputs of the sub-multiplication logics. In an embodiment of the disclosure, the activated shifting logic may perform left shifting on the outputs of the sub-multiplication logics by a predefined bit width. In an embodiment of the disclosure, the electronic device **3000** may obtain the output of the multiplication logic based on the shifted or unshifted outputs of the sub-multiplication logics.

[0262] FIG. **21** is a flowchart for explaining the operation of a selective adder according to an embodiment of the disclosure. Redundant descriptions given with reference to FIGS. **1** to **20** are omitted.

[0263] In an embodiment of the disclosure, operations S**2110** and S**2120** may be performed by at least one of the electronic device **3000**, the multiplier **3110**, the hardware accelerator **3100**, and the processor **3200** of FIG. **18**. However, the disclosure is not limited thereto, and operations S**2110** and S**2120** may be performed by any electronic device. An operating method of the selective adder or a controlling method of the selective adder according to an embodiment of the disclosure is not limited to that shown in FIG. **21**, may omit at least one of the operations shown in FIG. **21**, and may further include an operation not shown in FIG. **21**.

[0264] In an embodiment of the disclosure, operations S**2110** and S**2120** may be performed by the electronic device **3000** between operations S**1930** and S**1940** of FIG. **19**. However, the time at which operations S**2110** and S**2120** are performed is not limited thereto.

[0265] In operation S**2110**, the electronic device **3000** may determine whether to activate the selective adder of a multiplication logic based on identified datatypes and precisions. The electronic device **3000** may determine whether to activate the selective adder mapped to the identified datatypes and precisions. The electronic device **3000** may activate or deactivate the selective adder according to an identification result. For example, the electronic device **3000** may activate the selective adder when the identified datatypes and precisions are bfloat16. For example, the electronic device **3000** may deactivate the selective adder when the identified datatypes and precisions are FP8.

[0266] In operation S**2120**, the electronic device **3000** may determine to activate the selective adder and activate the selective adder that sums outputs of sub-multiplication logics. In an embodiment of the disclosure, the activated selective adder may sum all the outputs of the sub-multiplication logics.

[0267] FIG. **22** is a flowchart for explaining the operation of addition logic according to an embodiment of the disclosure. Redundant descriptions given with reference to FIGS. **1** to **21** are omitted.

[0268] In an embodiment of the disclosure, operations S**2210** and S**2220** may be performed by at least one of the electronic device **3000**, the multiplier **3110**, the hardware accelerator **3100**, and the processor **3200** of FIG. **18**. However, the disclosure is not limited thereto, and operations S**2210** and S**2220** may be performed by any electronic device. An operating method of the addition logic or a controlling method of the addition logic according to an embodiment of the disclosure is not limited to that shown in FIG. **22**, may omit at least one of the operations shown in FIG. **22**, and may further include an operation not shown in FIG. **22**.

[0269] In an embodiment of the disclosure, operations S**2210** and S**2220** may be performed by the electronic device **3000** after operation S**1940** of FIG. **19**. However, the time at which operations S**2210** and S**2220** are performed is not limited thereto.

[0270] In operation S**2210**, the electronic device **3000** may determine whether to activate the addition logic based on identified datatypes and precisions. In an embodiment of the disclosure, the electronic device **3000** may determine whether to activate at least one sub-addition logic of the addition logic.

[0271] In operation S**2220**, the electronic device **3000** may determine to activate the addition logic and activate the addition logic that sums bits corresponding to an exponent of a second input value and bits corresponding to an exponent of a first input value, and subtracts bias values mapped to the identified datatypes and precisions from a summation result. The at least one sub-addition logic of the activated addition logic may perform addition between exponents by using two adders. One adder may sum the bits corresponding to the exponent of the first input value and the bits corresponding to the exponent of the second input value. Another adder may subtract the bias values mapped to the identified datatypes and precisions from a summation result.

[0272] FIG. **23** is a flowchart for explaining the operation of XOR logic according to an embodiment of the disclosure. Redundant descriptions given with reference to FIGS. **1** to **22** are omitted.

[0273] In an embodiment of the disclosure, operations S**2310** and S**2320** may be performed by at least one of the electronic device **3000**, the multiplier **3110**, the hardware accelerator **3100**, and the processor **3200** of FIG. **18**. However, the disclosure is not limited thereto, and operations S**2310** and S**2320** may be performed by any electronic device. An operating method of the XOR logic or a controlling method of the XOR logic according to an embodiment of the disclosure is not limited to that shown in FIG. **23**, may omit at least one of the operations shown in FIG. **23**, and may further include an operation not shown in FIG. **23**.

[0274] In an embodiment of the disclosure, operations S**2310** and S**2320** may be performed by the electronic device **3000** after operation S**2220** of FIG. **22**. However, the time at which operations S**2310** and S**2320** are performed is not limited thereto.

[0275] In operation S**2310**, the electronic device **3000** may determine whether to activate the XOR logic based on identified datatypes and precisions. In an embodiment of the disclosure, the electronic device **3000** may determine whether to activate at least one sub-XOR logic of the XOR logic.

[0276] In operation S**2320**, the electronic device **3000** may determine to perform an XOR operation between a bit corresponding to a sign of a first input value and a bit corresponding to a sign of a second input value based on an activation of the XOR logic.

[0277] FIG. **24** is a flowchart for explaining the operations of a normalizer and a rounder according to an embodiment of the disclosure. Redundant descriptions given with reference to FIGS. **1** to **23** are omitted.

[0278] In an embodiment of the disclosure, operations S**2410** to S**2430** may be performed by at least one of the electronic device **3000**, the multiplier **3110**, the hardware accelerator **3100**, and the processor **3200** of FIG. **18**. However, the disclosure is not limited thereto, and operations S**2410** to S**2430** may be performed by any electronic device. An operating method of the normalizer and the rounder or a controlling method of the normalizer and the rounder according to an embodiment of the disclosure is not limited to that shown in FIG. **24**, may omit at least one of the operations shown in FIG. **24**, and may further include an operation not shown in FIG. **24**.

[0279] In an embodiment of the disclosure, operations S**2410** to S**2430** may be performed by the electronic device **3000** after operation S**2320** of FIG. **23**. However, the time at which operations S**2410** to S**2430** are performed is not limited thereto.

[0280] In operation S**2410**, the electronic device **3000** may determine whether to activate the normalizer and the rounder based on identified datatypes and precisions. In an embodiment of the disclosure, the normalizer may include at least one sub-normalizer. The rounder may include at least one sub-rounder. The number of activated sub-normalizers may be equal to the number of activated sub-rounders.

[0281] In operation S**2420**, the electronic device **3000** may determine to activate the normalizer and activate the normalizer that normalizes at least one output of a multiplication logic and at least one output of an addition logic. In an embodiment of the disclosure, the number of sub-normalizers activated may vary depending on the identified datatypes and precisions.

[0282] In operation S**2430**, the electronic device **3000** may determine to activate the rounder and activate the rounder that performs rounding with a predefined bit width by using at least one output of the XOR logic and at least one output of the normalizer. In an embodiment of the disclosure, the number of activated sub-rounders may vary depending on the identified datatypes and precisions.

[0283] According to an embodiment of the disclosure, operations on various precisions and datatypes may be supported while maintaining the accuracy of training or inference of the DNN model.

[0284] According to an embodiment of the disclosure, an area-efficient multiplier that supports various precisions and datatypes may be provided. One multiplier may perform multiplication operations not only on floating-point types but also on integer types, and thus, a large number of multipliers may be integrated per unit area. In other words, there is no need to have different multipliers with respect to different precisions and datatypes.

[0285] According to an embodiment of the disclosure, a flexible and general-purpose MAC array with high hardware resource utilization with respect to various operations of the DNN model may be provided. The MAC array according to an embodiment of the disclosure is designed to have high resource utilization for both accumulated and non-accumulated operations, and one hardware accelerator is designed to process a plurality of DNN models in parallel.

[0286] For example, an application that provides augmented reality or virtual reality service needs to simultaneously perform tasks such as object detection and depth estimation, and thus, to provide the service of the application, the hardware accelerator may process the plurality of DNN models corresponding to the tasks in parallel.

[0287] In an embodiment of the disclosure, an electronic device may be provided. The electronic device may include a multiplier including a multiplication logic. The electronic device may include a memory comprising at least one instruction. The electronic device may include at least one processor configured to execute the at least one instruction. The at least one processor may obtain a first input value and a second input value. The at least one processor may identify datatypes and precisions of the first input value and the second input value. The at least one processor may, based on the identified datatypes and precisions, distribute bits of the first input value and bits of the second input value to sub-multiplication logics of the multiplication logic. The at least one processor may obtain at least one output of the multiplication logic based on outputs of the sub-multiplication logics.

[0288] In an embodiment of the disclosure, the at least one processor may, based on the identified datatypes and precisions, determine whether to activate at least one shifting logic of the multiplication logic. The at least one processor may determine to activate the at least one shifting logic and activate the at least one shifting logic configured to shift at least one of the outputs of the sub-multiplication logics.

[0289] In an embodiment of the disclosure, the at least one processor may, based on the identified datatypes and precisions, determine whether to activate a selective adder of the multiplication logic. The at least one processor may determine to activate the selective adder and activate the selective adder configured to sum outputs of the sub-multiplication logics. The at least one processor may determine to deactivate the selective adder and deactivate the selective adder configured to sum the outputs of the sub-multiplication logics.

[0290] In an embodiment of the disclosure, when the identified datatypes are floating-point types, the bits of the first input value distributed to the sub-multiplication logics may be bits corresponding to a mantissa of the first input value.

[0291] In an embodiment of the disclosure, the bits of the second input value distributed to the sub-multiplication logics may be bits corresponding to a mantissa of the second input value.

[0292] In an embodiment of the disclosure, the multiplier may include an addition logic.

[0293] In an embodiment of the disclosure, the at least one processor may, based on the identified datatypes and precisions, determine whether to activate the addition logic. The at least one processor may determine to activate the addition logic and activate the addition logic configured to sum bits corresponding to an exponent of the second input value to bits corresponding to an exponent of the first input value and subtract bias values mapped to the identified datatypes and precisions from a summation result.

[0294] In an embodiment of the disclosure, the multiplier may include an XOR logic.

[0295] In an embodiment of the disclosure, the at least one processor may, based on the identified datatypes and precisions, determine whether to activate the XOR logic. The at least one processor may determine to activate the XOR logic and activate the XOR logic configured to perform an XOR operation between a bit corresponding to a sign of the first input value and a bit corresponding to a sign of the second input value.

[0296] In an embodiment of the disclosure, the XOR logic may include at least one sub-XOR logic.

[0297] In an embodiment of the disclosure, the at least one sub-XOR logic may be arranged in the same number as a maximum number of at least one output of the multiplication logic.

[0298] In an embodiment of the disclosure, the multiplier may include a normalizer.

[0299] In an embodiment of the disclosure, the multiplier may include a rounder.

[0300] According to an embodiment of the disclosure, the at least one processor may, based on the identified datatypes and precisions, determine whether to activate the normalizer

and the rounder. The at least one processor may determine to activate the normalizer and activate the normalizer configured to normalize at least one output of the multiplication logic and at least one output of the addition logic. The at least one processor may determine to activate the rounder and activate the rounder configured to perform rounding with a predefined bit width by using at least one output of the XOR logic and at least one output of the normalizer.

[0301] In an embodiment of the disclosure, the at least one processor may, based on the identified datatypes being integer types, determine at least one output of the multiplication logic to be an output of the multiplier. The at least one processor may, based on the identified datatypes being floating-point types, determine the at least one output of the normalizer as an output of the multiplier.

[0302] In an embodiment of the disclosure, an operating method of a multiplier may be provided. The method may include obtaining a first input value and a second input value. The method may include identifying datatypes and precisions of the first input value and the second input value. The method may include, based on the identified datatypes and precisions, distributing bits of the first input value and bits of the second input value to sub-multiplication logics of a multiplication logic included in the multiplier. The method may include obtaining at least one output of the multiplication logic based on outputs of the sub-multiplication logics.

[0303] In an embodiment of the disclosure, a multiplier may be provided. The multiplier may include a multiplication logic configured to, based on datatypes of a first input value and a second input value being integer types, perform a multiplication between the first input value and the second input value, and based on the datatypes being floating-point types, perform a multiplication between a mantissa of the first input value and a mantissa of the second input value. The multiplier may include an addition logic configured to, based on the datatypes being floating-point types, sum an exponent of the first input value and an exponent of the second input value, and subtract bias values mapped to bit widths of the exponent of the first input value and the exponent of the second input value from a summation result. The multiplier may include an XOR logic configured to, based on the datatypes being floating-point types, perform an XOR operation between a sign of the first input value and a sign of the second input value. The multiplier may include a normalizer configured to, based on the datatypes being floating-point types, perform normalization an output of the multiplication logic and an output of the addition logic. The multiplier may include a rounder configured to, based on the datatypes being floating-point types, perform rounding with a predefined bit width by using an output of the normalizer and an output of the XOR logic. The multiplier may include a multiplexer configured to, based on the datatypes being integer types, output the output of the multiplication logic, and based on the datatypes being floating-point types, output an output of the rounder.

[0304] In an embodiment of the disclosure, a hardware accelerator may be provided. The hardware accelerator may include a MAC array. The MAC array may include a plurality of arrays. The MAC array may include a global multiplexer logic connecting or disconnecting at least two of the plurality of arrays based on a first control signal. Each of the plurality of arrays may include a plurality of sub-arrays including a plurality of first sub-arrays and a second sub-array. Each of the plurality of arrays may include a local multiplexer logic connecting or disconnecting the plurality of sub-arrays based on a second control signal.

[0305] In an embodiment of the disclosure, the MAC array may perform a 128×128 matrix operation. Each of the plurality of arrays may may perform a 64×64 matrix operation.

[0306] In an embodiment of the disclosure, MAC operators may be arranged in each of the plurality of first sub-arrays in the form of 9×64. MAC operators may be arranged in the second sub-array in the form of 1×64.

[0307] In an embodiment of the disclosure, the number of plurality of arrays may be 4. The number of plurality of first sub-arrays may be 7.

[0308] In an embodiment of the disclosure, each of the MAC operators may include a multiplier that performs multiplication between an input value and a weight value. Each of the MAC operators may include an adder summing a partial sum value and an output of the multiplier. Each of the MAC operators may include a precision converter that converts precision of the output of the multiplier. Each of the MAC operators may include an adder tree summing the partial sum value and an output of the precision converter. Each of the MAC operators may include a first multiplexer that outputs an output of the adder or an output of the adder tree based on a third control signal.

[0309] In an embodiment of the disclosure, based on the first control signal indicating a first operation mode of the MAC array, the global multiplexer logic may disconnect the plurality of arrays from each other.

[0310] In an embodiment of the disclosure, based on the first control signal indicating a second operation mode of the MAC array, the global multiplexer logic may connect at least two of the plurality of arrays to each other.

[0311] In an embodiment of the disclosure, the plurality of arrays may be grouped into at least one array group by the global multiplexer logic. Each of the at least one array group may perform an operation on different single tenants.

[0312] In an embodiment of the disclosure, based on the second control signal indicating a first operation mode of the plurality of arrays, the local multiplexer logic may disconnect the plurality of sub-arrays from each other.

[0313] In an embodiment of the disclosure, based on the second control signal indicating a second operation mode of the plurality of arrays, the local multiplexer logic may connect the plurality of sub-arrays to each other.

[0314] In an embodiment of the disclosure, the plurality of sub-arrays may generate a partial sum tensor by performing a convolution operation between a first tensor and a second tensor. Based on the second control signal indicating a first operation mode of the plurality of arrays, the plurality of arrays may accumulate a partial sum tensor output by each of the plurality of sub-arrays. Based on the second control signal indicating a second operation mode of the plurality of arrays, the plurality of arrays may not accumulate the partial sum tensor output by each of the plurality of sub-arrays.

[0315] In an embodiment of the disclosure, a MAC array may be provided. The MAC array may include four arrays, each including seven first sub-arrays, one second sub-array, and local multiplexer logic. The MAC array may include global multiplexer logic disposed between the arrays. Each of the first sub-arrays may include a plurality of MAC operators arranged in the form of 9×64. The second sub-array may include a plurality of MAC operators arranged in

the form of 1×64. The local multiplexer logic may be disposed between the first sub-arrays and the second sub-array.

[0316] In an embodiment of the disclosure, the global multiplexer logic may include a plurality of first multiplexers connected between the plurality of MAC operators disposed in a last row, a last column, a first row, or a first column of the arrays.

[0317] In an embodiment of the disclosure, each of the plurality of first multiplexers may transfer an output (e.g., a partial sum or forwarded data) of one of the arrays to another array based on the first control signal corresponding to a first logic value. Each of the plurality of first multiplexers may not transfer the output (e.g., a partial sum or forwarded data) of one of the arrays to another array, based on the first control signal corresponding to a second logic value.

[0318] In an embodiment of the disclosure, the local multiplexer logic may include a plurality of second multiplexers connected to the plurality of MAC operators arranged in a last row or a last column of the first sub-arrays.

[0319] In an embodiment of the disclosure, each of the plurality of second multiplexers may transfer an output of one of the first sub-arrays to another first sub-array or the second sub-array based on the second control signal corresponding to the first logical value. Each of the plurality of second multiplexers may not transfer the output of one of the first sub-arrays to another first sub-array or the second sub-array based on the second control signal corresponding to the second logical value.

[0320] In an embodiment of the disclosure, when the second control signal corresponds to the first logic value, each of the arrays may perform a 64×64 matrix operation.

[0321] In an embodiment of the disclosure, values of the weight tensor corresponding to an input channel may be respectively prefetched to the arrays in a column direction.

[0322] Values of the weight tensor corresponding to an output channel may be respectively prefetched to the arrays in a row direction.

[0323] In an embodiment of the disclosure, when the second control signal corresponds to the second logic value, values of a weight filter of the weight tensor may be respectively prefetched to the first sub-arrays in the column direction. When the second control signal corresponds to the second logic value, the values of the weight tensor corresponding to the output channel may be respectively prefetched to the arrays in the row direction.

[0324] In an embodiment of the disclosure, when the second control signal corresponds to the second logic value, the MAC operators of the second sub-array may be grouped into seven groups of nine each.

[0325] In an embodiment of the disclosure, the values of the weight tensor may be respectively prefetched the seven groups.

[0326] A method according to an embodiment may be implemented in the form of program instructions executable by various computer means and be recorded on a computer-readable recording medium. The computer-readable recording medium may include program instructions, data files, data structures, or the like alone or in combination. The program instructions recorded on the computer-readable recording medium may be program instructions specially designed and configured for the disclosure or program instructions known and available to those of ordinary skill in the field of computer software. Examples of the computer-readable recording medium may include magnetic media such as a hard disk, a floppy disk, and magnetic tape, optical recording media such as compact disc read-only memory (CD-ROM) and a digital versatile disk (DVD), magneto-optical media such as a floptical disk, and hardware devices such as ROM, RAM, and flash memory, which are specially configured to store and execute program instructions. Examples of the program instructions may include machine language code produced by a compiler and high-level language code executable by a computer by using an interpreter or etc.

[0327] Some embodiments of the disclosure may be implemented in the form of a recording medium including instructions executable by a computer, such as program modules executed by a computer. The computer-readable recording medium may be any available medium accessible by a computer and includes volatile and non-volatile media and separable and non-separable media. In addition, the computer-readable recording medium may include a computer storage medium and a communication medium. The computer-readable recording medium includes volatile and non-volatile media and separable and non-separable media, which are implemented by any method or technique for storing information, such as computer-readable instructions, data structures, program modules, or other data. The communication medium typically includes computer-readable instructions, data structures, program modules, other data in modulated data signals such as carrier waves, or other transfer mechanisms, and includes any information transfer medium. In addition, some embodiments of the disclosure may be implemented in the form of a computer program or computer program product including instructions executable by a computer, such as a computer program executed by a computer

[0328] In an embodiment of the disclosure, a machine-readable storage medium may be provided in the form of a non-transitory storage medium. Here, the term "non-transitory storage medium" only means that the storage medium is tangible and does not include signals (for example, electromagnetic waves), whether data is semi permanently or temporarily stored in the storage medium or not. For example, the "non-transitory storage medium" may include a buffer in which data is temporarily stored.

[0329] According to an embodiment of the disclosure, a method according to various embodiments may be provided while included in a computer program product. The computer program product may be traded as merchandise between a seller and a purchaser. The computer program product may be distributed in the form of a machine-readable storage medium (for example, CD-ROM) or may be distributed (for example, downloaded or uploaded) online through an application store or directly between two user devices (for example, smartphones). In the case of online distribution, at least a portion of the computer program product (for example, a downloadable app) may be at least temporarily stored in a machine-readable storage medium, such as a memory of a server of a manufacturer, a server of an application store, or a relay server, or may be temporarily generated.

[0330] It should be understood that embodiments described herein should be considered in a descriptive sense only and not for purposes of limitation. Descriptions of features or aspects within each embodiment should typically be considered as available for other similar features or

aspects in other embodiments. While one or more embodiments have been described with reference to the figures, it will be understood by those of ordinary skill in the art that various changes in form and details may be made therein without departing from the spirit and scope of the disclosure as defined by the following claims.

What is claimed is:

1. An electronic device comprising:

a multiplier comprising a multiplication logic;

a memory storing at least one instruction; and

at least one processor configured to execute the at least one instruction to:

obtain a first input value and a second input value,

identify datatypes and precisions of the first input value and the second input value,

based on the identified datatypes and precisions, distribute bits of the first input value and bits of the second input value to sub-multiplication logics of the multiplication logic, and

obtain at least one output of the multiplication logic based on outputs of the sub-multiplication logics.

2. The electronic device of claim **1**, wherein

the at least one processor is configured to,

based on the identified datatypes and precisions, determine whether to activate at least one shifting logic of the multiplication logic, and

determine to activate the at least one shifting logic and activate the at least one shifting logic configured to shift at least one of the outputs of the sub-multiplication logics.

3. The electronic device of claim **1**, wherein the at least one processor is configured to, based on the identified datatypes and precisions, determine whether to activate a selective adder of the multiplication logic.

4. The electronic device of claim **3**, wherein

the at least one processor is configured to

determine to activate the selective adder and activate the selective adder configured to sum outputs of the sub-multiplication logics, and

determine to deactivate the selective adder and deactivate the selective adder configured to sum the outputs of the sub-multiplication logics.

5. The electronic device of claim **1**, wherein when the identified datatypes are floating-point types, the bits of the first input value distributed to the sub-multiplication logics are bits corresponding to a mantissa of the first input value, and the bits of the second input value distributed to the sub-multiplication logics are bits corresponding to a mantissa of the second input value.

6. The electronic device of claim **1**, wherein

the multiplier includes an addition logic, and

the at least one processor is configured to,

based on the identified datatypes and precisions, determine whether to activate the addition logic, and

determine to activate the addition logic and activate the addition logic configured to sum bits corresponding to an exponent of the second input value to bits corresponding to an exponent of the first input value and subtract bias values mapped to the identified datatypes and precisions from a summation result.

7. The electronic device of claim **6**, wherein

the multiplier includes an XOR logic, and

the at least one processor is configured to,

based on the identified datatypes and precisions, determine whether to activate the XOR logic, and

determine to activate the XOR logic and activate the XOR logic configured to perform an XOR operation between a bit corresponding to a sign of the first input value and a bit corresponding to a sign of the second input value.

8. The electronic device of claim **7**, wherein

the XOR logic includes at least one sub-XOR logic, and

the at least one sub-XOR logic is arranged in a same number as a maximum number of at least one output of the multiplication logic.

9. The electronic device of claim **7**, wherein

the multiplier includes a normalizer and a rounder, and

the at least one processor is configured to,

based on the identified datatypes and precisions, determine whether to activate the normalizer and the rounder,

determine to activate the normalizer and activate the normalizer configured to normalize at least one output of the multiplication logic and at least one output of the addition logic, and

determine to activate the rounder and activate the rounder configured to perform rounding with a predefined bit width by using at least one output of the XOR logic and at least one output of the normalizer.

10. The electronic device of claim **9**, wherein

the at least one processor is configured to,

based on the identified datatypes being integer types, determine at least one output of the multiplication logic to be an output of the multiplier, and

based on the identified datatypes being floating-point types, determine the at least one output of the normalizer as an output of the multiplier.

11. An operating method of a multiplier, the operating method comprising:

obtaining a first input value and a second input value,

identifying datatypes and precisions of the first input value and the second input value,

based on the identified datatypes and precisions, distributing bits of the first input value and bits of the second input value to sub-multiplication logics of a multiplication logic included in the multiplier, and

obtaining at least one output of the multiplication logic based on outputs of the sub-multiplication logics.

12. The operating method of claim **11**, further comprising:

based on the identified datatypes and precisions, determining whether to activate at least one shifting logic of the multiplication logic, and

determining to activate the at least one shifting logic and activating the at least one shifting logic configured to shift at least one of the outputs of the sub-multiplication logics.

13. The operating method of claim **11**, further comprising:

based on the identified datatypes and precisions, determining whether to activate a selective adder of the multiplication logic.

14. The operating method of claim **13**, wherein

the determining of whether to activate the selective adder of the multiplication logic includes

determining to activate the selective adder and activating the selective adder configured to sum outputs of the sub-multiplication logics; and

determining to deactivate the selective adder and deactivating the selective adder configured to sum the outputs of the sub-multiplication logics.

15. The operating method of claim **11**, wherein when the identified datatypes are floating-point types, the bits of the first input value distributed to the sub-multiplication logics are bits corresponding to a mantissa of the first input value, and the bits of the second input value distributed to the sub-multiplication logics are bits corresponding to a mantissa of the second input value.

16. The operating method of claim **11**, wherein

the multiplier includes an addition logic, and

the operating method further comprising:

based on the identified datatypes and precisions, determining whether to activate the addition logic; and

determining to activate the addition logic and activating the addition logic configured to sum bits corresponding to an exponent of the second input value to bits corresponding to an exponent of the first input value and subtract bias values mapped to the identified datatypes and precisions from a summation result.

17. The operating method of claim **16**, wherein

the multiplier includes an XOR logic,

the operating method further comprising:

based on the identified datatypes and precisions, determining whether to activate the XOR logic, and

determining to activate the XOR logic and activating the XOR logic configured to perform an XOR operation between a bit corresponding to a sign of the first input value and a bit corresponding to a sign of the second input value.

18. The operating method of claim **17**, wherein

the XOR logic includes at least one sub-XOR logic, and

the at least one sub-XOR logic is arranged in a same number as a maximum number of at least one output of the multiplication logic.

19. The operating method of claim **17**, wherein

the multiplier includes a normalizer and a rounder,

the operating method further comprising:

based on the identified datatypes and precisions, determining whether to activate the normalizer and the rounder,

determining to activate the normalizer and activating the normalizer configured to normalize at least one output of the multiplication logic and at least one output of the addition logic, and

determining to activate the rounder and activating the rounder configured to perform rounding with a predefined bit width by using at least one output of the XOR logic and at least one output of the normalizer.

20. A multiplier comprising:

a multiplication logic configured to, based on datatypes of a first input value and a second input value being integer types, perform a multiplication between the first input value and the second input value, and based on the datatypes being floating-point types, perform a multiplication between a mantissa of the first input value and a mantissa of the second input value;

an addition logic configured to, based on the datatypes being floating-point types, sum an exponent of the first input value and an exponent of the second input value, and subtract bias values mapped to bit widths of the exponent of the first input value and the exponent of the second input value from a summation result;

an XOR logic configured to, based on the datatypes being floating-point types, perform an XOR operation between a sign of the first input value and a sign of the second input value;

a normalizer configured to, based on the datatypes being floating-point types, perform normalization an output of the multiplication logic and an output of the addition logic;

a rounder configured to, based on the datatypes being floating-point types, perform rounding with a predefined bit width by using an output of the normalizer and an output of the XOR logic; and

a multiplexer configured to, based on the datatypes being integer types, output the output of the multiplication logic, and based on the datatypes being floating-point types, output an output of the rounder.

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